

Mutually Exclusive Courses

Courses of which a student may credit only one — for approval

Compiled on 1 October 2026

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1 How to use this booklet

1. *Looking up one course?* Use the Course Finder (Section 3, page 18). Every course code in this booklet has its own line there, with its equivalent codes, every course it cannot be credited together with, and links to the rows.
2. *Equivalent courses* (the first table of each school): one row for each course that exists under several codes. The primary (current) code is on the left and the other codes of the same course are on the right. A student may credit *only one code from the row*.
3. *Courses that cannot be credited together* (the second table of each school): different courses whose content overlaps. A student who has credited the course on the left may not credit the course on the right, and vice versa. The row also covers all equivalent codes of both courses.
4. In every table, entries already in the curriculum or course records (**Confirmed**) come first. Entries found by comparing syllabi (**Proposed**) follow, with the highest share of common content first.

1.1 Status of an entry

Confirmed	Already written in the curriculum, a course record, the DC Curriculum or the Minors booklet. The source is given in the row.
Proposed	Not written anywhere yet. Found by comparing the syllabi and checked by reading both; the share of common content is given. Needs approval.
Partly confirmed	An equivalence row in which some codes are confirmed and some are proposed. Each code shows its own status.

Sources: Master Course List, the course proposals in curriculum_pdfs, the DC Curriculum (batches 2023–26) and the Minors booklet. Appendices A–C are for the approving committee.

2 Mutually exclusive courses by school

2.1 Centre for Artificial Intelligence and Robotics (AR)

2.1.1 Equivalent courses

Only one code of each row may be credited.

Row	Primary course	Equivalent codes of the same course, with the reason	Status
E1	AR-501 Robot Kinematics, Dynamics, and Control (3-1-0-4)	ME-452 Robotics and Control (Robot Kinematics, Dynamics and Control) (2-1-0-3) — Minors booklet lists AR-501 / ME-452	Confirmed
E2	AR-520 Design Practicum of Mechatronic Systems (2-0-2-3)	AR-503 Mechatronics (3-0-0-3) — Minors booklet: 'equivalent courses'	Confirmed
E3	AR-514 Vision and Learning Based Control (3-0-0-3)	EE-543 Vision and Learning Based Control (3-0-0-3) — Proposed: Same syllabus (about 100%)	Proposed

2.1.2 Courses that cannot be credited together

Row	Course	Cannot be credited together with	Status	Common content	Source / main common topics
N1	AR-516 Introduction to Blockchain and Web3 (3-0-0-3)	MB-554 Blockchain for Business (2-0-0-2)	Proposed	60%	Both cover blockchain basics, consensus (PoW/PoS/BFT), Bitcoin mining/ transactions, Ethereum smart contracts, Hyperledger Fabric and use cases

2.2 Biosciences and Bioengineering (BE, BY)

2.2.1 Equivalent courses

Only one code of each row may be credited.

Row	Primary course	Equivalent codes of the same course, with the reason	Status
E4	BE-309 Biosensing and Bioinstrumentation (3-0-2-4)	BE-503 Biosensing and Bioinstrumentation (3-0-2-4) — BE-503 changed to BE-309 (49th BoA)	Confirmed
E5	BE-310 Biomaterials (3-0-2-4)	BE-504 Biomaterials (3-0-2-4) — BE-504 changed to BE-310 (49th BoA)	Confirmed
E6	BY-537 Computational Biology-1 (3-0-2-4)	BY-512 Quantitative and Computational Biology (3-0-0-3) — BY-512 replaced by BY-537	Confirmed

2.2.2 Courses that cannot be credited together

Row	Course	Cannot be credited together with	Status	Common content	Source / main common topics
N2	BY-537 Computational Biology-1 (3-0-2-4) and its equivalents (row E6): BY-512 Quantitative and Computational Biology	BY-606 Bioinformatics Applications for Systems Analysis (2-0-2-3)	Proposed	80%	All six BY-537 modules (programming, databases/file formats, sequence alignment, gene prediction, molecular phylogeny, systems biology) appear with ...
N3	BY-510 Advanced Cell Biology (3-0-0-3)	BY-530 Advanced Cell and Molecular Biology (3-0-0-3)	Proposed	75%	All BY-510 modules (cell tour, membrane transport, signalling, ER trafficking, cell cycle/apoptosis/cancer) reappear verbatim, condensed, in BY-530, ...
N4	BE-304 Bioinformatics (3-0-2-4)	BY-606 Bioinformatics Applications for Systems Analysis (2-0-2-3)	Proposed	50%	BE-304 (intro, Perl scripting, file formats, databases, pairwise/multiple alignment, BLAST/Clustal, motifs) is almost wholly inside BY-606 Parts I-II
N5	BE-310 Biomaterials (3-0-2-4) and its equivalents (row E5): BE-504 Biomaterials	MT-506 Biomaterials (3-0-0-3)	Proposed	50%	Shared: biomaterial properties/ biocompatibility, metal/ceramic/polymer classes, cell-material interaction, orthopaedic/dental implants

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Row	Course	Cannot be credited together with	Status	Common content	Source / main common topics
N6	BY-502 Biophysics and Protein Engineering (3-0-0-3)	BY-519 Protein Sciences in Therapeutics (3-0-0-3)	Proposed	50%	Verbatim shared modules: Intrinsically Disordered Proteins, Thermodynamics/protein folding, Protein Engineering basics (~22 of 42 lec)
N7	BY-507 Genetic Engineering (3-0-2-4)	BY-515 Molecular Biotechnology (3-0-0-3)	Proposed	50%	BY-515 repeats BY-507 gene-cloning principles, DNA manipulative enzymes/ligation and cloning vectors (lambda, M13, pUC8, cosmids) verbatim

2.3 Civil and Environmental Engineering (CE)

2.3.1 Equivalent courses

Only one code of each row may be credited.

Row	Primary course	Equivalent codes of the same course, with the reason	Status
E7	CE-203P Building Materials Lab (0-0-2-1)	CE-354P Building and Pavement Materials Laboratory (0-0-2-1) — DC curriculum lists CE-203P / CE-354P	Confirmed
E8	CE-310P Strength of Materials and Structures Lab (0-0-2-1)	CE-301P Strength of Materials and Structures Lab (0-0-2-1) — DC curriculum lists CE-301P / CE-310P	Confirmed
E9	CE-311P Geotechnical Engineering Laboratory (0-0-2-1)	CE-302P Geotechnical Engineering Lab (L-T-P-C not recorded) — DC curriculum lists CE-302P / CE-311P	Confirmed
E10	CE-251 Hydraulics Engineering (3-0-0-3)	ME-210 Fluid Mechanics (2.5-0.5-0-3) — Proposed: Same syllabus (about 90%)	Proposed
E11	CE-403 Wastewater Engineering (3-0-0-3)	CE-405 Water and Wastewater Engineering (3-0-0-3) — Proposed: Same syllabus (about 80%)	Proposed

2.3.2 Courses that cannot be credited together

Row	Course	Cannot be credited together with	Status	Common content	Source / main common topics
N8	CE-310 Strength of Materials and Structures (3-0-0-3)	CE-301 Strength of Materials and Structures (3-0-2-4)	Confirmed	75%	DC curriculum lists CE-301 / CE-310
N9	CE-311 Geotechnical Engineering I (3-0-0-3)	CE-302 Geotechnical Engineering (3-0-2-4)	Confirmed	35%	DC curriculum lists CE-302 / CE-311
N10	CE-301 Strength of Materials and Structures (3-0-2-4)	CE-310P Strength of Materials and Structures Lab (0-0-2-1) and its equivalents (row E8): CE-301P Strength of Materials and Structures Lab	Proposed	95%	CE-310P's experiments (beam reactions, SF/BM, truss/portal deflection, arches, strut buckling, torsion) are the lab already inside CE-301 (3-0-2-4)
N11	CE-302 Geotechnical Engineering (3-0-2-4)	CE-402 Geotechnical Engineering II (2-1-0-3)	Proposed	90%	Every CE-402 module (subsoil exploration, earth pressure, bearing capacity, shallow foundations, settlement, piles) is near-verbatim in CE-302 (30 of ...)
N12	CE-404 Analysis of Structures (3-0-0-3)	CE-506 Analysis of Indeterminate Structures (3-0-0-3)	Proposed	65%	Shared: indeterminacy, force method/least work, moment distribution, three-moment theorem, portal/cantilever approx. analysis, intro matrix methods
N13	CE-402 Geotechnical Engineering II (2-1-0-3)	CE-513 Advanced Foundation Engineering (3-0-0-3)	Proposed	40%	Shared: Terzaghi bearing capacity, shallow foundation choice/expansive soils, immediate and consolidation settlement, pile capacity/groups/negative ...

Courses of this school listed under other schools: CE-101 Engineering Graphics → E64; CE-102 Environmental Studies → E65; CE-530 Continuum Mechanics → E83, N72; CE-562 Numerical Methods for Engineering Computation → E80; CE-563 Finite Element Methods in Engineering → E86; CE-564 Fundamentals of Fracture Mechanics → E81; CE-607 FEM (discontinued, 24th Senate) → E86; CE-617 Mechanics of Composite Materials → E84.

2.4 Computer Science and Engineering (CS)

2.4.1 Equivalent courses

Only one code of each row may be credited.

Row	Primary course	Equivalent codes of the same course, with the reason	Status
E12	CS-207 Applied Databases Practicum (0-0-3-2)	MA-323 Applied Databases Practicum (0-0-3-2) — Stated in MA-323 (59th BoA)	Confirmed
E13	CS-212 Design and Analysis of Algorithms (3-0-2-4)	MA-312 Design and Analysis of Algorithms (3-0-2-4) — Stated in MA-312 (59th BoA); DC lists MA-312 / CS-212	Confirmed
E14	CS-304 Formal Languages and Automata Theory (3-0-0-3)	MA-313 Formal Languages and Automata Theory (3-0-0-3) — Stated in MA-313 (59th BoA); DC lists MA-313 / CS-304	Confirmed
E15	CS-315 Artificial Intelligence (3-0-2-4)	CS-305 Artificial Intelligence (3-0-0-3) — Recoded w.e.f. B25 (DC curriculum) CS-362 Artificial Intelligence (3-0-0-3) — DC curriculum: CS-315 (CS-362, CS-562) CS-562 Artificial Intelligence (3-0-0-3) — DC curriculum: CS-315 (CS-362, CS-562)	Confirmed
E16	CS-514 Data Structures and Algorithms-II (3-0-2-4)	CS-580 Advanced Data Structures and Algorithms (old) (3-0-1-4) — DC curriculum: CS-514 (CS-580)	Confirmed
E17	CS-520 Introduction to Quantum Computing (2-0-1-3)	QT-405 Introduction to Quantum Computation (3-0-0-3) — Minors booklet lists QT-405 / CS-520	Confirmed
E18	CS-671 Deep Learning and Applications (3-1-0-4)	DS-417 Deep Learning (3-0-2-4) — DC curriculum: DSAI-only code for CS-671	Confirmed
E19	CS-683 Generative AI (3-0-1-4)	DS-418 Introduction to Generative AI (3-0-2-4) — DC curriculum: DSAI-only code for CS-683	Confirmed
E20	CS-312 Operating System (3-0-2-4)	CS-306 Operating Systems (3-0-0-3) — Proposed: Same syllabus (about 90%)	Proposed
E21	CS-302 Paradigms of Programming (3-0-2-4)	CS-205 Paradigms of Programming (3-0-0-3) — Proposed: Same syllabus (about 85%)	Proposed
E22	CS-603 Managerial Decision Making (3-0-0-3)	HS-616 Managerial Thinking and Decision Making (3-0-0-3) — Proposed: Same syllabus (about 80%)	Proposed
E23	CS-313 Computer Networks (3-0-2-4)	CS-206 Computer Networks (3-0-0-3) — Proposed: Same syllabus (about 80%) CS-406 Computer Networks (3-1-0-4) — Proposed: Same syllabus (about 75%)	Proposed
E24	CS-214 Computer Organization (3-0-2-4)	CS-201 Computer Organization (3-0-0-3) — Proposed: Same title; syllabus not on record EE-306 Computer Organization & Microprocessor (3-0-0-3) — Proposed: Same syllabus (about 85%) VL-326 Computer Organization (3-0-2-4) — Proposed: Same title; syllabus not on record	Proposed
E25	CS-501 Access Networks (3-0-0-3)	EE-533 Access Network (4 cr.) — Proposed: Same title; syllabus not on record	Proposed

2.4.2 Courses that cannot be credited together

Row	Course	Cannot be credited together with	Status	Common content	Source / main common topics
N14	CS-212 Design and Analysis of Algorithms (3-0-2-4) and its equivalents (row E13): MA-312 Design and Analysis of Algorithms	CS-403 Algorithm Design and Analysis (3-0-2-4)	Confirmed	—	Stated in MA-312 (59th BoA)
N15	CS-511 Applied Probability (2-0-0-2)	EE-534 Probability and Random Processes (3-0-0-3) and its equivalents (row E38): CS-601 Probability and Random Processes	Confirmed	80%	Stated in CS-511 (44th BoA)
N16	(same course as above)	IC-210 Probability, Statistics and Random Processes (2.5-0.5-0-3)	Confirmed	65%	CS-511 proposal: >30% overlap declared
N17	(same course as above)	MA-524 Probability and Statistics (3-1-0-4)	Confirmed	—	Stated in CS-511 (44th BoA)
N18	CS-512 Matrix Theory (2-0-0-2)	EE-522 Matrix Theory for Engineers (3-0-0-3)	Confirmed	65%	Stated in CS-512 (44th BoA)
N19	(same course as above)	IC-114 Linear Algebra (2-0-0-2) and its equivalents (row E66): IC-111 Linear Algebra	Confirmed	—	CS-512 proposal: >30% overlap declared
N20	(same course as above)	MA-512 Linear Algebra (3-1-0-4)	Confirmed	—	Stated in CS-512 (44th BoA)

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Row	Course	Cannot be credited together with	Status	Common content	Source / main common topics
N21	CS-513 Discrete Mathematics (3-1-0-4)	CS-208 Mathematical Foundations of Computer Science (3-1-0-4)	Confirmed	—	Stated in CS-513 (44th BoA)
N22	(same course as above)	CS-511 Applied Probability (2-0-0-2)	Confirmed	—	Stated in CS-513
N23	CS-514 Data Structures and Algorithms-II (3-0-2-4) and its equivalents (row E16): CS-580 Advanced Data Structures and Algorithms (old)	CS-403 Algorithm Design and Analysis (3-0-2-4)	Confirmed	—	CS-514 is only for students who have not taken CS-403
N24	CS-520 Introduction to Quantum Computing (2-0-1-3) and its equivalents (row E17): QT-405 Introduction to Quantum Computation	PH-579 Quantum Computation and Information (3-0-0-3)	Proposed	60%	Shared qubits/postulates, quantum gates/circuits, no-cloning, teleportation, Deutsch-Jozsa, Simon, Grover, Shor, BB84
N25	CS-202 Data Structures and Algorithms (3-1-0-4)	IC-253 Programming and Data Structures (2-0-2-3)	Proposed	55%	IC-253 repeats CS-202 text on complexity, stacks/queues, lists, trees, BST/AVL, heaps, graph traversal
N26	CS-212 Design and Analysis of Algorithms (3-0-2-4) and its equivalents (row E13): MA-312 Design and Analysis of Algorithms	CS-514 Data Structures and Algorithms-II (3-0-2-4) and its equivalents (row E16): CS-580 Advanced Data Structures and Algorithms (old)	Proposed	50%	Shared: sorting review, amortized analysis, DP, Bellman-Ford/max-flow, union-find/quake heaps, NP-completeness, approx/FPT, convex hull, 28h coding ...
N27	CS-303 Software Engineering (2-0-2-3)	CS-347 Software Engineering (3-0-2-4)	Proposed	50%	Same title
N28	CS-609 Speech Processing (3-0-2-4)	CS-630 Speech Technology (3-0-2-4)	Proposed	50%	Shared speech production, DSP review, short-time spectral analysis/spectrograms, LPC, cepstrum, HMM speech recognition, speaker recognition and ...

Courses of this school listed under other schools: CS-508 Introduction to Heterogeneous Computing → N89; CS-511 Applied Probability → N33, N46; CS-512 Matrix Theory → N41; CS-601 Probability and Random Processes → E38, N34, N47, N57, N60; CS-669 Pattern Recognition → E35, N82.

2.5 Chemical Sciences (CY)

2.5.1 Equivalent courses

Only one code of each row may be credited.

Row	Primary course	Equivalent codes of the same course, with the reason	Status
E26	CY-001 Preparatory Chemistry - 1 (3-1-0-4)	CY-100 Preparatory Chemistry 1 (3-0-0-3) — Proposed: Same title; syllabus not on record	Proposed
E27	CY-531P Organic Chemistry Laboratory (3 cr.)	CY-501P Organic Chemistry Laboratory (0-0-8-4) — Proposed: Same title; syllabus not on record	Proposed
E28	CY-532 Photochemistry and Pericyclic Reactions (3 cr.)	CY-502 Photochemistry and pericyclic reactions (3-1-0-4) — Proposed: Same title; syllabus not on record	Proposed
E29	CY-533 Chemistry of Main Group Elements (3 cr.)	CY-507 Chemistry of Main Group Elements (3-1-0-4) — Proposed: Same title; syllabus not on record	Proposed
E30	CY-533P Inorganic Chemistry Lab (0-0-8-4)	CY-506P Inorganic Chemistry Lab (0-0-8-4) — Proposed: Same title; syllabus not on record	Proposed
E31	CY-534 Chemistry of Transition Elements (3-1-0-4)	CY-506 Chemistry of Transition Elements (3-1-0-4) — Proposed: Same title; syllabus not on record	Proposed

2.5.2 Courses that cannot be credited together

Row	Course	Cannot be credited together with	Status	Common content	Source / main common topics
N29	CY-241 Nano-scale Science and Technology (3-0-0-3)	CY-554 Science and Technology of Nanomaterials (3-0-0-3)	Proposed	75%	Same structure and textbooks: nanoscale properties, synthesis, SPM/SEM/TEM, metal/semiconductor/CNT/graphene/fullerene nanomaterials, applications ...
N30	CY-515 Advanced Inorganic Spectroscopy (3-0-0-3)	CY-553 Organic Inorganic Spectroscopy (3-0-0-3)	Proposed	55%	Shared EPR, NQR and Mossbauer modules (–23 of 42 h
N31	CY-523 Colloids and Interface Science and Technology (3-0-0-3)	MT-510 Colloids and Interfaces (3-0-0-3)	Proposed	50%	Shared: colloid classification and stability (DLVO), van der Waals forces/ Hamaker/medium effects, electrical double layer, contact angle and wetting, ...
N32	CY-511 Group Theory and Spectroscopy (3-0-0-3)	CY-701 Advances Physical Methods in Chemistry Theory and Applications (3-0-0-3)	Proposed	40%	Shared group theory/symmetry of vibrations, light-matter interaction (Einstein coefficients, Beer-Lambert, Franck-Condon) and rotational/ vibrational ...

Courses of this school listed under other schools: CY-552 Hydrogen Generation and Storage → N97.

2.6 Data Science (DS)

2.6.1 Equivalent courses

Only one code of each row may be credited.

Row	Primary course	Equivalent codes of the same course, with the reason	Status
E32	DS-314 Probability Theory and Statistics (3-1-0-4)	DS-303 Statistical Foundations of Data Science (3-0-0-3) — Proposed: Same syllabus (about 100%) DS-313 Statistical Foundations of Data Science (3-1-0-4) — Recoded w.e.f. B25 (DC curriculum)	Partly confirmed
E33	DS-312 Computing Systems for Data Processing (3-0-2-4)	DS-302 Computing Systems for Data Processing (2-0-2-3) — Recoded w.e.f. B25 (DC curriculum) MA-120 Introduction to Computing System and Databases (3-0-2-4) — Proposed: Same syllabus (about 85%)	Partly confirmed
E34	DS-415 Optimization for Machine Learning (3-0-2-4)	DS-401 Optimization for Data Science (3-0-0-3) — Proposed: Same syllabus (about 85%) DS-411 Optimization for Data Science (3-1-0-4) — Recoded w.e.f. B25 (DC curriculum)	Partly confirmed
E35	DS-416 Fundamentals of Machine Learning (3-0-2-4)	CS-669 Pattern Recognition (3-1-0-4) — DC curriculum: DS-416 (CS-669) DS-403 Introduction to Statistical Learning (2-0-2-3) — Proposed: Same syllabus (about 85%) DS-413 Introduction to Statistical Learning (3-1-0-4) — Recoded w.e.f. B25 (DC curriculum)	Partly confirmed
E36	DS-412 Matrix Computations for Data Science (3-0-2-4)	DS-402 Matrix Computations for Data Science (2-0-2-3) — Proposed: Same syllabus (about 90%)	Proposed

2.6.2 Courses that cannot be credited together

Row	Course	Cannot be credited together with	Status	Common content	Source / main common topics
N33	DS-314 Probability Theory and Statistics (3-1-0-4) and its equivalents (row E32): DS-303 Statistical Foundations of Data Science DS-313 Statistical Foundations of Data Science	CS-511 Applied Probability (2-0-0-2)	Confirmed	60%	Stated in DS-313

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Row	Course	Cannot be credited together with	Status	Common content	Source / main common topics
N34	(same course as above)	EE-534 Probability and Random Processes (3-0-0-3) and its equivalents (row E38): CS-601 Probability and Random Processes	Confirmed	60%	Stated in DS-313
N35	(same course as above)	MA-524 Probability and Statistics (3-1-0-4)	Confirmed	—	Stated in DS-313
N36	(same course as above)	MB-510 Probability and Statistics for Data Science and AI (2-0-0-2)	Confirmed	—	Stated in DS-313
N37	DS-415 Optimization for Machine Learning (3-0-2-4) and its equivalents (row E34): DS-401 Optimization for Data Science DS-411 Optimization for Data Science	EE-530 Applied Optimization (3-0-0-3)	Proposed	65%	Both follow Boyd: convex sets/cones, convex functions/conjugate, LP/QP/GP/SOCP/SDP, Lagrange duality/KKT
N38	DS-314 Probability Theory and Statistics (3-1-0-4) and its equivalents (row E32): DS-303 Statistical Foundations of Data Science DS-313 Statistical Foundations of Data Science	IC-210 Probability, Statistics and Random Processes (2.5-0.5-0-3)	Proposed	55%	Shared: axioms/Bayes, RVs/pmf/pdf/joint/conditional, MGF/characteristic functions, Chebyshev/CLT, autocorrelation/stationarity/ergodicity, Markov ...
N39	DS-415 Optimization for Machine Learning (3-0-2-4) and its equivalents (row E34): DS-401 Optimization for Data Science DS-411 Optimization for Data Science	MA-576 Nonlinear Optimization (3-0-2-4)	Proposed	50%	Shared: convex sets/functions, optimality conditions, Lagrange duality, gradient descent, Newton, barrier/interior-point

Courses of this school listed under other schools: DS-402 Matrix Computations for Data Science → N79; DS-403 Introduction to Statistical Learning → N82; DS-412 Matrix Computations for Data Science → N79; DS-413 Introduction to Statistical Learning → N82; DS-416 Fundamentals of Machine Learning → N82; DS-417 Deep Learning → E18; DS-418 Introduction to Generative AI → E19.

2.7 Electrical Engineering and VLSI (EE, VL)

2.7.1 Equivalent courses

Only one code of each row may be credited.

Row	Primary course	Equivalent codes of the same course, with the reason	Status
E37	EE-314 Digital Signal Processing (3-0-2-4)	EE-305 Digital Signal Processing (3-1-0-4) — Minors booklet: 'equivalent – only one counts'	Confirmed
E38	EE-534 Probability and Random Processes (3-0-0-3)	CS-601 Probability and Random Processes (3-1-0-4) — EE-534 proposal: replaces CS-601	Confirmed
E39	EE-261 Electrical Systems Around Us (3-0-0-3)	EE-101 Electrical Science (3-1-2-4) — 'Equivalent Course: IC 160' in EE-101 IC-160 Electrical Systems Around Us (2.5-0.5-0-3) — Proposed: Same title; syllabus not on record	Partly confirmed
E40	EE-518 Information Theory (3-0-0-3)	EE-605 Information Theory (3 cr.) — Proposed: Same syllabus (about 100%)	Proposed
E41	EE-607 Optical Communication Systems (3-0-0-3)	EE-614 Optical Communication Systems (3-0-0-3) — Proposed: Same syllabus (about 100%)	Proposed
E42	EE-231 Measurement and Instrumentation (2-0-2-3)	IC-231 Measurement and Instrumentation (2-0-2-3) — Proposed: Same syllabus (about 75%)	Proposed
E43	EE-570 Advanced Communication Theory (3-0-2-4)	EE-503 Advance Communication Theory (3-0-0-3) — Proposed: Same syllabus (about 65%)	Proposed
E44	EE-302 Control Systems (3-0-2-4)	EE-301 Control Systems (old version) (2.5-0.5-0-3) — Proposed: Same title; syllabus not on record EE-602 Control System Applications (2.5-0.5-0-3) — Proposed: Same syllabus (about 95%)	Proposed
E45	EE-212 Digital System Design (3-0-2-4)	EE-210 Digital System Design (3-0-0-3) — Proposed: Same title; syllabus not on record	Proposed
E46	EE-260 Signals and Systems (2.5-0.5-0-3)	IC-260 Signals and Systems (2.5-0.5-0-3) — Proposed: Same title; syllabus not on record	Proposed
E47	EE-304 Communication Systems (3-0-1-4)	EE-316 Communication Systems (3-0-2-4) — Proposed: Same title; syllabus not on record	Proposed

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Row	Primary course	Equivalent codes of the same course, with the reason	Status
E48	EE-512 CMOS Analog IC Design (3-0-2-4)	VL-404 CMOS Analog IC Design (3-0-2-4) — Proposed: Same title; syllabus not on record	Proposed
E49	EE-555 Intelligent Control System (3-0-0-3)	EE-597 Intelligent Control System (3-0-0-3) — Proposed: Same title; syllabus not on record	Proposed
E50	EE-601 Advanced Electric Drives (2.5-0.5-0-3)	EE-604 Advanced Electric Drives (3 cr.) — Proposed: Same title; syllabus not on record	Proposed

2.7.2 Courses that cannot be credited together

Row	Course	Cannot be credited together with	Status	Common content	Source / main common topics
N40	EE-205 Electromagnetics and Wave Propagation (3-0-0-3)	PH-521 Electromagnetic Theory (4-0-0-4) and its equivalents (row E95): PH-421 Electromagnetic Theory	Confirmed	25%	Stated in EE-205 (52nd BoA)
N41	EE-553 Foundations of Intelligent Communication Systems-I (3-0-2-4)	CS-512 Matrix Theory (2-0-0-2)	Confirmed	—	Stated in EE-553 (54th BoA)
N42	(same course as above)	EE-522 Matrix Theory for Engineers (3-0-0-3)	Confirmed	—	Stated in EE-553 (54th BoA)
N43	(same course as above)	MA-221 Numerical Analysis (3-1-0-4) and its equivalents (row E75): MA-523 Numerical Analysis MA-551 Numerical Analysis MA-607 Numerical Analysis	Confirmed	—	Stated in EE-553 (54th BoA)
N44	(same course as above)	MA-512 Linear Algebra (3-1-0-4)	Confirmed	—	Stated in EE-553 (54th BoA)
N45	(same course as above)	MA-525 Heuristic Optimization (3-0-0-3)	Confirmed	—	Stated in EE-553 (54th BoA)
N46	EE-581 Applied Statistics for Data and Signals (3-0-2-4)	CS-511 Applied Probability (2-0-0-2)	Confirmed	—	Stated in EE-581 (54th BoA)
N47	(same course as above)	EE-534 Probability and Random Processes (3-0-0-3) and its equivalents (row E38): CS-601 Probability and Random Processes	Confirmed	—	Stated in EE-581 (54th BoA)
N48	(same course as above)	MA-511 Real Analysis (3-1-0-4)	Confirmed	—	Stated in EE-581 (54th BoA)
N49	(same course as above)	MA-521 Functional Analysis (3-1-0-4)	Confirmed	—	Stated in EE-581 (54th BoA)
N50	(same course as above)	MA-524 Probability and Statistics (3-1-0-4)	Confirmed	—	Stated in EE-581 (54th BoA)
N51	(same course as above)	MA-568 Real Analysis (2.5-0.5-0-3)	Confirmed	—	Stated in EE-581 (54th BoA)
N52	(same course as above)	MA-601 Real and Functional Analysis (2-1-0-3)	Confirmed	—	Stated in EE-581 (54th BoA)
N53	EE-616 Microwave engineering (2-0-0-2)	EE-622 Microwave Integrated Circuits (3-0-0-3)	Proposed	85%	Near-verbatim shared modules: S-parameters/matching, Wilkinson/hybrid couplers, image-parameter/insertion-loss filters, ferrite isolators, transistor ...
N54	EE-524 Digital MOS LSI Circuits (3-0-0-3)	EE-678 Digital Circuit Design (3 cr.)	Proposed	80%	Same Rabaey-based sequence: MOS review, CMOS inverter, static/dynamic/pass-transistor logic, latches/registers/pipelining, clocking/skew/PLL, ...
N55	EE-303 Power Systems (3-1-0-4)	EE-552 Power and Energy Systems (3-1-0-4)	Proposed	70%	Near-verbatim shared: system structure, component representation/line parameters/per unit, NR load flow, symmetrical faults, ELD, ...
N56	EE-531 Estimation and Detection Theory (3-0-0-3)	EE-575 Applied Statistics for Data and Signals (3-0-2-4)	Proposed	70%	EE-575 copies EE-531 detection theory (NP, ROC, GLRT, LMP), MVU/CRLB, BLUE/ML/LS and Bayesian MMSE/ MAP/Wiener modules

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Row	Course	Cannot be credited together with	Status	Common content	Source / main common topics
N57	EE-534 Probability and Random Processes (3-0-0-3) and its equivalents (row E38): CS-601 Probability and Random Processes	IC-210 Probability, Statistics and Random Processes (2.5-0.5-0-3)	Proposed	65%	Shared: axioms/Bayes, RVs pdf/cdf, moments/MGF/Chebyshev/CLT, joint distributions, random processes (autocorrelation, stationarity, white noise)
N58	EE-211 Analog Circuit Design (2-0-2-3)	EE-677 Analog Circuit Design (3 cr.)	Proposed	55%	Shared: single/two-transistor amplifiers, current mirrors/active loads, differential pairs, output stages, frequency response, feedback and stability
N59	EE-311 Device Electronics for Integrated Circuits (3-0-0-3)	EE-520 Microelectronics Devices and Modelling (3-0-0-3)	Proposed	55%	Shared: semiconductor physics/band model/carriers, p-n junctions and breakdown, metal-semiconductor contacts, MOS capacitor/MOSFET and short channel
N60	EE-534 Probability and Random Processes (3-0-0-3) and its equivalents (row E38): CS-601 Probability and Random Processes	MA-524 Probability and Statistics (3-1-0-4)	Proposed	50%	Shared: axioms, sigma fields/measure, conditional probability/Bayes, RVs/pdf/cdf, standard distributions, moments/MGF, CLT, ...
N61	EE-231 Measurement and Instrumentation (2-0-2-3) and its equivalents (row E42): IC-231 Measurement and Instrumentation	EE-313 Measurement and Instrumentation (2-1-0-3)	Proposed	40%	Same title in EE
N62	EE-508 Fundamentals of Electric Drives (3-0-0-3)	EE-601 Advanced Electric Drives (2.5-0.5-0-3) and its equivalents (row E50): EE-604 Advanced Electric Drives	Proposed	40%	Shared: drive/load-torque basics and multi-quadrant operation, closed-loop DC drives, scalar/vector control of IM, self-controlled synchronous, ...
N63	EE-575 Applied Statistics for Data and Signals (3-0-2-4)	EE-581 Applied Statistics for Data and Signals (3-0-2-4)	Proposed	35%	Borderline

Courses of this school listed under other schools: EE-306 Computer Organization & Microprocessor → E24; EE-522 Matrix Theory for Engineers → N18; EE-530 Applied Optimization → N37; EE-533 Access Network → E25; EE-534 Probability and Random Processes → N15, N34; EE-543 Vision and Learning Based Control → E3; VL-326 Computer Organization → E24.

2.8 Energy Engineering and Electric Transportation (EN, EM, ET)

2.8.1 Equivalent courses

Only one code of each row may be credited.

Row	Primary course	Equivalent codes of the same course, with the reason	Status
E51	EM-504 Materials for Energy Applications (3-1-0-4)	EM-601 (title not recorded) (L-T-P-C not recorded) — EN list of equivalent courses (14th Senate)	Confirmed
E52	EN-509 Functional Materials for Energy Engineering (3-0-0-3)	ME-502 (old) Functional Materials (old ME-502) (3-0-0-3) — Old ME-502 changed to ME-609/EN-509 ME-609 Functional Materials (3-0-0-3) — Stated in EN-509; ME-609 was modified into it (75%) MT-205 Functional Properties of Materials (3-0-2-4) — Stated in MT-205 (57th BoA)	Confirmed
E53	EN-611 Durability Behavior of Energy Materials (3-0-0-3)	EM-658 Deformation Behavior of Energy Materials (3-0-0-3) — EN list of equivalent courses (18th Senate)	Confirmed
E54	EN-504 Energy: Environment Policy and Law (3-0-0-3)	EM-604 Energy: Environment Policy and Law (3-1-0-4) — Proposed: Same syllabus (about 100%) EN-604 Energy: Environment Policy and Law (3-0-0-3) — Proposed: Same syllabus (about 100%)	Proposed

2.8.2 Courses that cannot be credited together

Row	Course	Cannot be credited together with	Status	Common content	Source / main common topics
N64	EN-605 Modeling and Simulation of Energy Systems (2-0-2-3)	ME-620 Modeling and Simulation (2-0-2-3)	Proposed	75%	Shared: dimensional analysis/similarity, lumped/distributed and state-space/transfer-function models, fuzzy inference, ANN backprop, GA/simulated ...
N65	EN-611 Durability Behavior of Energy Materials (3-0-0-3) and its equivalents (row E53): EM-658 Deformation Behavior of Energy Materials	MT-304 Mechanical Behavior of Materials (3-0-2-4)	Proposed	65%	MT-304 modules 1-4 are verbatim EN-611 (static deformation, dislocations/strengthening, creep, fatigue)
N66	EN-512 Structure-Property Correlations for Energy Applications (1-0-4-3)	EN-612 Structure - Property correlation in materials for Energy Applications (2-0-2-3)	Proposed	40%	Near-identical titles

Courses of this school listed under other schools: EN-503 Energy Storage Technologies → N96; EN-506 Design of Energy Systems → E91; EN-509 (old) Thermodynamics for Energy Systems (old EN-509) → E85, N90, N93; EN-512 Structure-Property Correlations for Energy Applications → N95.

2.9 General Engineering (GE)

2.9.1 Equivalent courses

Only one code of each row may be credited.

Row	Primary course	Equivalent codes of the same course, with the reason	Status
E55	GE-521 Essentials of Entrepreneurship (3-0-0-3)	HS-510 Essentials of Entrepreneurship (3-0-0-3) — Proposed: Same syllabus (about 100%)	Proposed

2.10 Humanities and Social Sciences (HS)

2.10.1 Equivalent courses

Only one code of each row may be credited.

Row	Primary course	Equivalent codes of the same course, with the reason	Status
E56	HS-307 Macroeconomics I (3-0-0-3)	MB-203 Macroeconomics (3 cr.) — Minors booklet: only one of HS-307 / MB-203	Confirmed
E57	HS-358 Science Writing (3-0-0-3)	HS-305 Science Writing (3-0-0-3) — Minors booklet lists HS-305 / HS-358	Confirmed
E58	HS-364 Modern China: A Historical Survey (3-0-0-3)	HS-392 Modern China (3-0-0-3) — 'Equivalent Course: HS 364' in HS-392	Confirmed
E59	HS-537 Post-Reform India: Polity, Society and Economy (3-0-0-3)	HS-471 Contemporary India (3-0-0-3) — HS-537 proposal: 50% overlap; HS-471 dropped (23rd Senate)	Confirmed
E60	HS-253 Introduction to Sociology (3-0-0-3)	HS-344 Introduction to Sociology (3-0-0-3) — Proposed: Same syllabus (about 100%)	Proposed
E61	HS-450 Financial Management (3-0-0-3)	HS-551 Financial Management (3-0-0-3) — Proposed: Same syllabus (about 90%)	Proposed
E62	HS-550 Statistical Methods (3-0-2-4)	HS-650 Statistical Methods (3-0-0-3) — Proposed: Same syllabus (about 90%)	Proposed
E63	HS-205 Financial Accounting (3-0-0-3)	MB-303 Financial Accounting (3 cr.) — Proposed: Same title; syllabus not on record	Proposed

2.10.2 Courses that cannot be credited together

Row	Course	Cannot be credited together with	Status	Common content	Source / main common topics
N67	HS-461 Consumer Behavior (3-0-0-3)	MB-582 Consumer Behavior (2-0-0-2)	Proposed	85%	Same Schiffman chapter sequence: segmentation, motivation/personality/perception/learning/attitudes, communication, family and culture, ...
N68	HS-382 Social Movements in India (3-0-0-3)	HS-536 Social Movements in India (3-0-0-3)	Proposed	60%	Identical title
N69	HS-450 Financial Management (3-0-0-3) and its equivalents (row E61): HS-551 Financial Management	MB-527 Financial Management (2-0-0-2)	Proposed	60%	Shared corporate-finance core: goal of firm/agency, time value of money, bond and stock valuation, capital investment NPV/cash flows
N70	HS-403 Organizational Behavior (3-0-0-3)	MB-524 Organizational Behaviour (2-0-0-2)	Proposed	50%	Same subject

Courses of this school listed under other schools: HS-304 Organizational Management → N88; HS-510 Essentials of Entrepreneurship → E55; HS-550 Statistical Methods → N83; HS-616 Managerial Thinking and Decision Making → E22; HS-650 Statistical Methods → N83.

2.11 Institute Core (IC)

2.11.1 Equivalent courses

Only one code of each row may be credited.

Row	Primary course	Equivalent codes of the same course, with the reason	Status
E64	IC-140 Engineering Graphics for Design (2-0-3-4)	CE-101 Engineering Graphics (2-0-4-4) — 'Equivalent Course: IC 140' in CE-101	Confirmed
E65	IC-230 Environmental Science (3-0-0-3)	CE-102 Environmental Studies (2-0-0-2) — 'Equivalent Course: IC 230' in CE-102	Confirmed
E66	IC-114 Linear Algebra (2-0-0-2)	IC-111 Linear Algebra (2.5-0.5-0-3) — Proposed: Same syllabus (about 100%)	Proposed
E67	IC-241 Materials Science for Engineers (3-0-0-3)	ME-204 Materials Science for Engineers (3-0-0-3) — Proposed: Same syllabus (about 75%)	Proposed

2.11.2 Courses that cannot be credited together

Row	Course	Cannot be credited together with	Status	Common content	Source / main common topics
N71	IC-110 Engineering Mathematics (2.5-0.5-0-3)	IC-112 Calculus (1.5-0.5-0-2)	Proposed	95%	IC-112 Calculus (single-variable calculus, several variables, infinite series) is wholly contained in IC-110, which also covers ODEs and double ...
N72	IC-242 Continuum Mechanics (2.5-0.5-0-3)	ME-530 Continuum Mechanics (3-0-0-3) and its equivalents (row E83): CE-530 Continuum Mechanics	Proposed	75%	Shared: Cartesian tensors, Cauchy stress/principal/deviatoric, Lagrangian/Eulerian kinematics and strain tensors, conservation laws with ...
N73	IC-110 Engineering Mathematics (2.5-0.5-0-3)	MA-101 Mathematics-I (3-1-0-4)	Proposed	70%	IC-110's several-variable calculus, infinite series tests and ODE modules appear verbatim in MA-101
N74	IC-112 Calculus (1.5-0.5-0-2)	MA-101 Mathematics-I (3-1-0-4)	Proposed	70%	Functions of several variables (Euler, Jacobians, Taylor, Lagrange multipliers) and series tests are verbatim in both
N75	IC-114 Linear Algebra (2-0-0-2) and its equivalents (row E66): IC-111 Linear Algebra	MA-512 Linear Algebra (3-1-0-4)	Proposed	65%	IC-114 almost contained in MA-512: systems/rank, eigenvalues and diagonalization, Hermitian/unitary/normal, vector spaces/basis/direct sum, linear ...
N76	IC-210 Probability, Statistics and Random Processes (2.5-0.5-0-3)	MA-524 Probability and Statistics (3-1-0-4)	Proposed	65%	Shared probability axioms/Bayes, RVs/cdf/pmf/pdf, same special-distributions list, CLT, estimation/CI, regression, hypothesis testing, ...

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Row	Course	Cannot be credited together with	Status	Common content	Source / main common topics
N77	IC-130 Applied Chemistry for Engineers (3-0-0-3)	IC-131 Applied Chemistry for Engineers (2-0-2-3)	Proposed	55%	Same title

Courses of this school listed under other schools: IC-111 Linear Algebra → N19; IC-114 Linear Algebra → N19; IC-141 Product Realization Technology → E89; IC-142 Engineering Thermodynamics → E90, N91, N93; IC-160 Electrical Systems Around Us → E39; IC-210 Probability, Statistics and Random Processes → N16, N38, N57; IC-221 Foundations of Electrodynamics → E98; IC-231 Measurement and Instrumentation → E42, N61; IC-253 Programming and Data Structures → N25; IC-260 Signals and Systems → E46.

2.12 Indian Knowledge System and Mental Health Applications (IK)

2.12.1 Equivalent courses

Only one code of each row may be credited.

Row	Primary course	Equivalent codes of the same course, with the reason	Status
E68	IK-101 Sanskrit Language – Level I (UG) (L-T-P-C not recorded)	IK-527 Sanskrit Language – Level I (PG) (3-0-0-3) — Stated in IK-527 (64th BoA)	Confirmed
E69	IK-504 Bhagavad Gita Comprehensive (3-0-0-3)	IK-530 Bhagavad-Gītā Part I (3-0-0-3) — Proposed: Same syllabus (about 100%)	Proposed
E70	IK-517 Philosophy of Science (3-0-0-3)	IK-593 Philosophy of Science (3 cr.) — Proposed: Same title; syllabus not on record	Proposed

2.13 Mathematical and Statistical Sciences (MA)

2.13.1 Equivalent courses

Only one code of each row may be credited.

Row	Primary course	Equivalent codes of the same course, with the reason	Status
E71	MA-211 Ordinary Differential Equations (3-1-0-4)	MA-513 Ordinary Differential Equations (3-1-0-4) — Stated in MA-211	Confirmed
E72	MA-220 Partial Differential Equations (3-1-0-4)	MA-522 Partial Differential Equations (3-1-0-4) — Stated in MA-220 (59th BoA)	Confirmed
E73	MA-560 Nonlinear Dynamics and Chaos (3-0-0-3)	MA-460 Nonlinear Dynamics and Chaos (3-0-0-3) — MA-560 is the revised MA-460	Confirmed
E74	MA-251 Abstract Algebra (3-0-0-3)	MA-548 Abstract Algebra (3-1-0-4) — Stated in both MA-251 and MA-548 (59th BoA) MA-549 Abstract Algebra (3-0-0-3) — Proposed: Same syllabus (about 85%)	Partly confirmed
E75	MA-221 Numerical Analysis (3-1-0-4)	MA-523 Numerical Analysis (3-1-0-4) — Stated in MA-221 (59th BoA) MA-551 Numerical Analysis (3-0-0-3) — Proposed: Same syllabus (about 90%) MA-607 Numerical Analysis (2-0-2-3) — Proposed: About 60% common content	Partly confirmed
E76	MA-550 Statistical Data Analysis (2-1-0-3)	MA-605 Statistical Data Analysis (2-1-0-3) — Proposed: Same syllabus (about 100%)	Proposed
E77	MA-780 Topics on Semigroup Theory (3-0-0-3)	MA-665 Semigroup Of Bounded Linear Operators (3 cr.) — Proposed: Same syllabus (about 100%)	Proposed
E78	MA-641 Operations Research (3-0-0-3)	MB-402 Operations Research (3 cr.) — Proposed: Same title; syllabus not on record	Proposed

2.13.2 Courses that cannot be credited together

Row	Course	Cannot be credited together with	Status	Common content	Source / main common topics
N78	MA-222 Applied Linear programming (3-1-0-4)	MA-515 Applied Mathematical Programming (3-1-0-4)	Confirmed	—	Stated in MA-222 (59th BoA)
N79	MA-310 Matrix Computations & Lab (3-0-2-4)	DS-412 Matrix Computations for Data Science (3-0-2-4) and its equivalents (row E36): DS-402 Matrix Computations for Data Science	Confirmed	—	Stated in MA-310 (59th BoA)

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Row	Course	Cannot be credited together with	Status	Common content	Source / main common topics
N80	MA-321 Numerics of Differential Equations (3-0-2-4)	MA-609 Numerics of Partial Differential Equations (3-0-0-3)	Confirmed	50%	Stated in MA-321 (59th BoA)
N81	MA-322 Applied Graph Theory (3-0-2-4)	MA-530 Graph Theory (3-1-0-4)	Confirmed	65%	Stated in MA-322 (59th BoA)
N82	MA-574 An Introduction to Probabilistic Machine Learning (3-0-2-4)	DS-416 Fundamentals of Machine Learning (3-0-2-4) and its equivalents (row E35): CS-669 Pattern Recognition DS-403 Introduction to Statistical Learning DS-413 Introduction to Statistical Learning	Confirmed	45%	Stated in MA-574 (59th BoA); Stated in MA-574 (DS-413, now DS-416)
N83	MA-611 Statistical tools and Computing (3-1-0-4)	HS-550 Statistical Methods (3-0-2-4) and its equivalents (row E62): HS-650 Statistical Methods	Confirmed	—	Stated in MA-611 (50th BoA)
N84	MA-210 Real and Complex Analysis (2.5-0.5-0-3)	MA-511 Real Analysis (3-1-0-4)	Proposed	65%	Shared: real number completeness/countability, R_n topology/ Bolzano-Weierstrass/compactness, metric spaces/continuity/connectedness, Riemann integral
N85	MA-511 Real Analysis (3-1-0-4)	MA-568 Real Analysis (2.5-0.5-0-3)	Proposed	65%	Same title
N86	MA-211 Ordinary Differential Equations (3-1-0-4) and its equivalents (row E71): MA-513 Ordinary Differential Equations	MA-465 Ordinary Differential Equations (3-0-0-3)	Proposed	60%	Shared: existence/uniqueness/continuation/dependence, fundamental matrix and stability of linear systems, Lyapunov, Sturm theory
N87	MA-521 Functional Analysis (3-1-0-4)	MA-601 Real and Functional Analysis (2-1-0-3)	Proposed	50%	MA-601's functional-analysis half (normed/Banach spaces, Riesz lemma, operator norm, Hahn-Banach, UBP, closed graph, open mapping, inner product ...)

Courses of this school listed under other schools: MA-101 Mathematics-I → N73, N74; MA-120 Introduction to Computing System and Databases → E33; MA-221 Numerical Analysis → N43; MA-312 Design and Analysis of Algorithms → E13, N14, N26; MA-313 Formal Languages and Automata Theory → E14; MA-323 Applied Databases Practicum → E12; MA-511 Real Analysis → N48; MA-512 Linear Algebra → N20, N44, N75; MA-521 Functional Analysis → N49; MA-523 Numerical Analysis → N43; MA-524 Probability and Statistics → N17, N35, N50, N60, N76; MA-525 Heuristic Optimization → N45; MA-551 Numerical Analysis → N43; MA-568 Real Analysis → N51; MA-576 Nonlinear Optimization → N39; MA-601 Real and Functional Analysis → N52; MA-607 Numerical Analysis → N43.

2.14 Management (MB)

2.14.1 Courses that cannot be credited together

Row	Course	Cannot be credited together with	Status	Common content	Source / main common topics
N88	MB-201 Foundations of Business Management (3-0-1-4)	HS-304 Organizational Management (3-0-0-3)	Confirmed	20%	Stated in MB-201 (54th BoA)

Courses of this school listed under other schools: MB-203 Macroeconomics → E56; MB-303 Financial Accounting → E63; MB-402 Operations Research → E78; MB-510 Probability and Statistics for Data Science and AI → N36; MB-524 Organizational Behaviour → N70; MB-527 Financial Management → N69; MB-554 Blockchain for Business → N1; MB-582 Consumer Behavior → N67.

2.15 Mechanical Engineering (ME, MI)

2.15.1 Equivalent courses

Only one code of each row may be credited.

Row	Primary course	Equivalent codes of the same course, with the reason	Status
E79	ME-304 Power Plant Engineering / Principles of Energy Conversion (3-0-2-4)	ME-356 Principles of Energy Conversion (3-1-0-4) — Stated in ME-304: 'equivalent to ME 356'	Confirmed
E80	ME-504 Numerical Methods for Engineering Computation (3-0-0-3)	CE-562 Numerical Methods for Engineering Computation (3-0-0-3) — Stated in CE-562 (59th BoA)	Confirmed
E81	ME-506 Fundamentals of Fracture Mechanics (2.5-0.5-0-3)	CE-564 Fundamentals of Fracture Mechanics (2.5-0.5-0-3) — Stated in CE-564 (59th BoA)	Confirmed
E82	ME-509 Nanomanufacturing (3-0-0-0)	ME-502 (old) Nanomanufacturing (old ME-502, 8th Senate) (3-0-0-3) — Old ME-502 changed to ME-509	Confirmed
E83	ME-530 Continuum Mechanics (3-0-0-3)	CE-530 Continuum Mechanics (3-0-0-3) — Stated in both records: 'CE 530 and ME 530 are same'	Confirmed
E84	ME-617 Mechanics of Composite Materials (3-0-0-3)	CE-617 Mechanics of Composite Materials (3-0-0-3) — Stated in CE-617 (59th BoA)	Confirmed
E85	ME-634 Thermodynamics for Energy Systems (3-0-0-3)	EN-509 (old) Thermodynamics for Energy Systems (old EN-509) (3-0-0-3) — Old EN-509 changed to ME-634 (11th Senate)	Confirmed
E86	ME-513 Finite Element Methods in Engineering (3-0-2-4)	CE-563 Finite Element Methods in Engineering (3-0-2-4) — Stated in CE-563 (59th BoA) CE-607 FEM (discontinued, 24th Senate) (L-T-P-C not recorded) — CE-607 discontinued (24th Senate); stated in CE-563 ME-352 Finite Element Methods in Engineering (2.5-0.5-0-3) — Proposed: Same syllabus (about 90%) ME-641 Finite Element Method (3-0-0-3) — Proposed: Same syllabus (about 85%)	Partly confirmed
E87	ME-501 Materials Science for Failure Analysis (3-0-0-3)	ME-607 Materials Science For Failure Analysis (3-0-0-3) — Proposed: Same syllabus (about 100%)	Proposed
E88	ME-311P Design Lab - 1 (0-0-2-1)	ME-306P Solid Mechanics Laboratory (3-0-0-3) — Proposed: Same syllabus (about 95%)	Proposed
E89	ME-212 Product Manufacturing Technology (2-0-3-3)	IC-141 Product Realization Technology (2-0-0-2) — Proposed: Same syllabus (about 90%)	Proposed
E90	ME-213 Engineering Thermodynamics (3-1-0-4)	IC-142 Engineering Thermodynamics (2.5-0.5-0-3) — Proposed: Same syllabus (about 85%)	Proposed
E91	ME-633 Design of Energy Systems (3-0-0-3)	EN-506 Design of Energy Systems (3-0-0-3) — Proposed: Same syllabus (about 80%)	Proposed

2.15.2 Courses that cannot be credited together

Row	Course	Cannot be credited together with	Status	Common content	Source / main common topics
N89	ME-522 High-Performance Scientific Computing (2-0-1-3)	CS-508 Introduction to Heterogeneous Computing (2-0-0-2)	Confirmed	—	Stated in ME-522 (46th BoA)
N90	ME-610 Advanced Thermodynamics (3-1-0-4)	ME-634 Thermodynamics for Energy Systems (3-0-0-3) and its equivalents (row E85): EN-509 (old) Thermodynamics for Energy Systems (old EN-509)	Proposed	85%	ME-634 modules (laws/LTE, entropy generation, Maxwell/Gibbs-Duhem multiphase, combustion, advanced power cycles) are near-verbatim in ME-610, which ...
N91	ME-213 Engineering Thermodynamics (3-1-0-4) and its equivalents (row E90): IC-142 Engineering Thermodynamics	MI-101 Thermodynamics (2-1-0-3)	Proposed	70%	MI-101 fully within IC-142: basic concepts, pure-substance properties, first law closed/open, second law/entropy, Otto/Diesel/Rankine and ...
N92	ME-309P Theory of Machines Lab (0-0-2-1)	ME-312P Design Lab - 2 (0-0-2-1)	Proposed	65%	Same experiments: mechanism synthesis, engine balancing, governors, gyroscope, cams, whirling of shaft, forced/torsional vibrations, transmissibility
N93	ME-213 Engineering Thermodynamics (3-1-0-4) and its equivalents (row E90): IC-142 Engineering Thermodynamics	ME-634 Thermodynamics for Energy Systems (3-0-0-3) and its equivalents (row E85): EN-509 (old) Thermodynamics for Energy Systems (old EN-509)	Proposed	45%	Shared: first law for open/closed and steady/transient, second law/entropy balance, Maxwell and Clausius-Clapeyron relations, exergy, power/combined ...

continued on the next page

Row	Course	Cannot be credited together with	Status	Common content	Source / main common topics
N94	ME-308 Manufacturing Engineering (3-0-0-3)	ME-315 Manufacturing Engineering-II (3-0-0-3)	Proposed	40%	Despite the II naming, shared rapid prototyping (FDM, LOM, SLA

Courses of this school listed under other schools: **ME-204** Materials Science for Engineers → [E67](#); **ME-210** Fluid Mechanics → [E10](#); **ME-452** Robotics and Control (Robot Kinematics, Dynamics and Control) → [E1](#); **ME-502 (old)** Functional Materials (old ME-502) → [E52](#); **ME-530** Continuum Mechanics → [N72](#); **ME-609** Functional Materials → [E52](#); **ME-620** Modeling and Simulation → [N64](#).

2.16 Materials Science and Engineering (MT)

2.16.1 Courses that cannot be credited together

Row	Course	Cannot be credited together with	Status	Common content	Source / main common topics
N95	MT-203 Materials Synthesis & Characterization (3-0-2-4)	EN-512 Structure-Property Correlations for Energy Applications (1-0-4-3)	Confirmed	—	Stated in MT-203 (57th BoA)
N96	MT-501 Energy Conversion & Storage Technology (3-0-0-3)	EN-503 Energy Storage Technologies (3-0-0-3)	Confirmed	75%	Stated in MT-501 (57th BoA)
N97	MT-509 Hydrogen Energy (3-0-0-3)	CY-552 Hydrogen Generation and Storage (3-0-0-3)	Confirmed	70%	Stated in MT-509 (57th BoA)

Courses of this school listed under other schools: **MT-205** Functional Properties of Materials → [E52](#); **MT-304** Mechanical Behavior of Materials → [N65](#); **MT-506** Biomaterials → [N5](#); **MT-510** Colloids and Interfaces → [N31](#).

2.17 Physical Sciences and Quantum Technologies (PH, EP, QS, QT)

2.17.1 Equivalent courses

Only one code of each row may be credited.

Row	Primary course	Equivalent codes of the same course, with the reason	Status
E92	PH-301 Quantum mechanics and applications (2.5-0.5-0-3)	QS-302 Foundation of Quantum Technology (3-0-0-3) — DC curriculum lists QS-302 / PH-301	Confirmed
E93	PH-550 Introduction to Quantum Optics (3-0-0-3)	QT-511 Introduction to Quantum Optics (3-0-0-3) — Minors booklet lists QT-511 / PH-550	Confirmed
E94	PH-611P Experimental Research Techniques (0-0-7-4)	PH-614P Experimental Research Techniques (0-0-7-4) — Proposed: Same syllabus (about 100%)	Proposed
E95	PH-521 Electromagnetic Theory (4-0-0-4)	PH-421 Electromagnetic Theory (4-0-0-4) — Proposed: Same syllabus (about 90%)	Proposed
E96	PH-612 Nuclear and Particle Physics (3-0-0-3)	PH-611 Nuclear and Particle Physics (4-0-0-4) — Proposed: Same syllabus (about 85%)	Proposed
E97	PH-302 Introduction to Statistical Mechanics (2.5-0.5-0-3)	PH-422 Statistical Mechanics (4-0-0-4) — Proposed: About 70% common content PH-522 Statistical Mechanics (4-0-0-4) — Proposed: Same syllabus (about 85%)	Proposed
E98	EP-321 Foundations of Electrodynamics (3-0-0-3)	IC-221 Foundations of Electrodynamics (2.5-0.5-0-3) — Proposed: Same title; syllabus not on record	Proposed

2.17.2 Courses that cannot be credited together

Row	Course	Cannot be credited together with	Status	Common content	Source / main common topics
N98	EP-403 Physics of atoms and molecules (3-0-0-3)	PH-524 Atomic and Molecular Physics (3-0-0-3)	Confirmed	90%	EP-403 proposal: 'cannot take these two courses'
N99	PH-301 Quantum mechanics and applications (2.5-0.5-0-3) and its equivalents (row E92): QS-302 Foundation of Quantum Technology	PH-513 Quantum Mechanics I (3-0-0-3)	Proposed	80%	Near-verbatim shared: postulates/ Ehrenfest, 1-D problems/tunneling, Dirac formalism, angular momentum, hydrogen atom, time-indep and time-dep ...

continued on the next page

Row	Course	Cannot be credited together with	Status	Common content	Source / main common topics
N100	EP-301 Engineering Mathematics-2 (3-1-0-4)	PH-511 Mathematical Physics (4-0-0-4)	Proposed	75%	EP-301 modules on vector spaces/ Hermitian operators, Frobenius/ Sturm-Liouville ODEs, special functions and complex analysis are verbatim in PH-511
N101	PH-424 Atomic and Molecular Spectroscopy (4-0-0-4)	PH-524 Atomic and Molecular Physics (3-0-0-3)	Proposed	75%	PH-524 content (perturbation theory, Rabi, Einstein coefficients, two-electron/ alkali atoms, LS/JJ, LCAO/BO, microwave/IR/Raman, NMR/ESR) is all in ...
N102	PH-613 Quantum Mechanics II (4-0-0-4)	PH-702 Advanced Quantum Mechanics (3-0-0-3)	Proposed	65%	Shared QM review, angular momentum/ CG, tensor operators/Wigner-Eckart, symmetries/parity/time reversal, second quantization, Klein-Gordon/Dirac
N103	EP-403 Physics of atoms and molecules (3-0-0-3)	PH-424 Atomic and Molecular Spectroscopy (4-0-0-4)	Proposed	60%	Shared perturbation theory, radiation-atom interaction (Rabi, Einstein coefficients, line broadening), two-electron atoms, LS/JJ coupling, ...

Courses of this school listed under other schools: PH-421 Electromagnetic Theory → N40; PH-521 Electromagnetic Theory → N40; PH-579 Quantum Computation and Information → N24; QT-405 Introduction to Quantum Computation → E17, N24.

2.18 All schools: Research Methodology courses

Only one of these courses may be credited. The rule is stated in PH-600 (“RM-600 in other schools”), MA-600, ME-600 and HC-600. HS-522 (4 credits) and IK-562 (3 credits) are full-length courses and are not part of this group.

Row	Primary course	Equivalent codes of the same course, with the reason	Status
E99	PH-600 Research Methodology (1-0-0-1)	BY-600 Research Methodology (1-0-0-1) — RM course of another school CE-600 Research Methodology for Civil Engineering (1-0-0-1) — RM course of another school CY-600 Research Methodology for Chemistry (1-0-0-1) — RM course of another school HC-600 Research Methodology (1-0-0-1) — RM course of another school HS-600 Research Methodology (1-0-0-1) — RM course of another school IK-509 Research methodology (1-0-0-1) — RM course of another school MA-600 Research Methodology (1-0-0-1) — RM course of another school ME-600 Research Methodology (1-0-0-1) — RM course of another school RM-510 Research Methodology (1-0-0-1) — RM course of another school RM-600 Research Methodology (1-0-0-1) — RM course of another school SC-600 Research Methodology (1 cr.) — RM course of another school	Confirmed

3 Course Finder

Every course code in this booklet, in code order. The row numbers are links. If a course is not listed here, no mutual exclusion is recorded or proposed for it.

Course (code, title, L-T-P-C)	Equivalent codes (only one may be credited)	Cannot be credited together with	Rows
AR-501 Robot Kinematics, Dynamics, and Control (3-1-0-4)	ME-452 Robotics and Control (Robot Kinematics, Dynamics and Control)	—	E1 (p. 3)
AR-503 Mechatronics (3-0-0-3)	AR-520 Design Practicum of Mechatronic Systems	—	E2 (p. 3)
AR-514 Vision and Learning Based Control (3-0-0-3)	EE-543 Vision and Learning Based Control	—	E3 (p. 3)
AR-516 Introduction to Blockchain and Web3 (3-0-0-3)	—	MB-554 Blockchain for Business	N1 (p. 3)
AR-520 Design Practicum of Mechatronic Systems (2-0-2-3)	AR-503 Mechatronics	—	E2 (p. 3)
BE-304 Bioinformatics (3-0-2-4)	—	BY-606 Bioinformatics Applications for Systems Analysis	N4 (p. 3)
BE-309 Biosensing and Bioinstrumentation (3-0-2-4)	BE-503 Biosensing and Bioinstrumentation	—	E4 (p. 3)
BE-310 Biomaterials (3-0-2-4)	BE-504 Biomaterials	MT-506 Biomaterials	E5 (p. 3) N5 (p. 3)
BE-503 Biosensing and Bioinstrumentation (3-0-2-4)	BE-309 Biosensing and Bioinstrumentation	—	E4 (p. 3)
BE-504 Biomaterials (3-0-2-4)	BE-310 Biomaterials	MT-506 Biomaterials	E5 (p. 3) N5 (p. 3)
BY-502 Biophysics and Protein Engineering (3-0-0-3)	—	BY-519 Protein Sciences in Therapeutics	N6 (p. 4)
BY-507 Genetic Engineering (3-0-2-4)	—	BY-515 Molecular Biotechnology	N7 (p. 4)
BY-510 Advanced Cell Biology (3-0-0-3)	—	BY-530 Advanced Cell and Molecular Biology	N3 (p. 3)
BY-512 Quantitative and Computational Biology (3-0-0-3)	BY-537 Computational Biology-1	BY-606 Bioinformatics Applications for Systems Analysis	E6 (p. 3) N2 (p. 3)
BY-515 Molecular Biotechnology (3-0-0-3)	—	BY-507 Genetic Engineering	N7 (p. 4)
BY-519 Protein Sciences in Therapeutics (3-0-0-3)	—	BY-502 Biophysics and Protein Engineering	N6 (p. 4)
BY-530 Advanced Cell and Molecular Biology (3-0-0-3)	—	BY-510 Advanced Cell Biology	N3 (p. 3)
BY-537 Computational Biology-1 (3-0-2-4)	BY-512 Quantitative and Computational Biology	BY-606 Bioinformatics Applications for Systems Analysis	E6 (p. 3) N2 (p. 3)
BY-600 Research Methodology (1-0-0-1)	CE-600 Research Methodology for Civil Engineering CY-600 Research Methodology for Chemistry HC-600 Research Methodology HS-600 Research Methodology IK-509 Research methodology MA-600 Research Methodology ME-600 Research Methodology PH-600 Research Methodology RM-510 Research Methodology RM-600 Research Methodology SC-600 Research Methodology	—	E99 (p. 17)
BY-606 Bioinformatics Applications for Systems Analysis (2-0-2-3)	—	BE-304 Bioinformatics BY-512 Quantitative and Computational Biology BY-537 Computational Biology-1	N2 (p. 3) N4 (p. 3)
CE-101 Engineering Graphics (2-0-4-4)	IC-140 Engineering Graphics for Design	—	E64 (p. 12)
CE-102 Environmental Studies (2-0-0-2)	IC-230 Environmental Science	—	E65 (p. 12)
CE-203P Building Materials Lab (0-0-2-1)	CE-354P Building and Pavement Materials Laboratory	—	E7 (p. 4)
CE-251 Hydraulics Engineering (3-0-0-3)	ME-210 Fluid Mechanics	—	E10 (p. 4)

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Course (code, title, L-T-P-C)	Equivalent codes (only one may be credited)	Cannot be credited together with	Rows
CE-301 Strength of Materials and Structures (3-0-2-4)	—	CE-301P Strength of Materials and Structures Lab CE-310 Strength of Materials and Structures CE-310P Strength of Materials and Structures Lab	N8 (p. 4) N10 (p. 4)
CE-301P Strength of Materials and Structures Lab (0-0-2-1)	CE-310P Strength of Materials and Structures Lab	CE-301 Strength of Materials and Structures	E8 (p. 4) N10 (p. 4)
CE-302 Geotechnical Engineering (3-0-2-4)	—	CE-311 Geotechnical Engineering I CE-402 Geotechnical Engineering II	N9 (p. 4) N11 (p. 4)
CE-302P Geotechnical Engineering Lab (L-T-P-C not recorded)	CE-311P Geotechnical Engineering Laboratory	—	E9 (p. 4)
CE-310 Strength of Materials and Structures (3-0-0-3)	—	CE-301 Strength of Materials and Structures	N8 (p. 4)
CE-310P Strength of Materials and Structures Lab (0-0-2-1)	CE-301P Strength of Materials and Structures Lab	CE-301 Strength of Materials and Structures	E8 (p. 4) N10 (p. 4)
CE-311 Geotechnical Engineering I (3-0-0-3)	—	CE-302 Geotechnical Engineering	N9 (p. 4)
CE-311P Geotechnical Engineering Laboratory (0-0-2-1)	CE-302P Geotechnical Engineering Lab	—	E9 (p. 4)
CE-354P Building and Pavement Materials Laboratory (0-0-2-1)	CE-203P Building Materials Lab	—	E7 (p. 4)
CE-402 Geotechnical Engineering II (2-1-0-3)	—	CE-302 Geotechnical Engineering CE-513 Advanced Foundation Engineering	N11 (p. 4) N13 (p. 4)
CE-403 Wastewater Engineering (3-0-0-3)	CE-405 Water and Wastewater Engineering	—	E11 (p. 4)
CE-404 Analysis of Structures (3-0-0-3)	—	CE-506 Analysis of Indeterminate Structures	N12 (p. 4)
CE-405 Water and Wastewater Engineering (3-0-0-3)	CE-403 Wastewater Engineering	—	E11 (p. 4)
CE-506 Analysis of Indeterminate Structures (3-0-0-3)	—	CE-404 Analysis of Structures	N12 (p. 4)
CE-513 Advanced Foundation Engineering (3-0-0-3)	—	CE-402 Geotechnical Engineering II	N13 (p. 4)
CE-530 Continuum Mechanics (3-0-0-3)	ME-530 Continuum Mechanics	IC-242 Continuum Mechanics	E83 (p. 15) N72 (p. 12)
CE-562 Numerical Methods for Engineering Computation (3-0-0-3)	ME-504 Numerical Methods for Engineering Computation	—	E80 (p. 15)
CE-563 Finite Element Methods in Engineering (3-0-2-4)	CE-607 FEM (discontinued, 24th Senate) ME-352 Finite Element Methods in Engineering ME-513 Finite Element Methods in Engineering ME-641 Finite Element Method	—	E86 (p. 15)
CE-564 Fundamentals of Fracture Mechanics (2.5-0.5-0-3)	ME-506 Fundamentals of Fracture Mechanics	—	E81 (p. 15)
CE-600 Research Methodology for Civil Engineering (1-0-0-1)	BY-600 Research Methodology CY-600 Research Methodology for Chemistry HC-600 Research Methodology HS-600 Research Methodology IK-509 Research methodology MA-600 Research Methodology ME-600 Research Methodology PH-600 Research Methodology RM-510 Research Methodology RM-600 Research Methodology SC-600 Research Methodology	—	E99 (p. 17)
CE-607 FEM (discontinued, 24th Senate) (L-T-P-C not recorded)	CE-563 Finite Element Methods in Engineering ME-352 Finite Element Methods in Engineering ME-513 Finite Element Methods in Engineering ME-641 Finite Element Method	—	E86 (p. 15)

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Course (code, title, L-T-P-C)	Equivalent codes (only one may be credited)	Cannot be credited together with	Rows
CE-617 Mechanics of Composite Materials (3-0-0-3)	ME-617 Mechanics of Composite Materials	—	E84 (p. 15)
CS-201 Computer Organization (3-0-0-3)	CS-214 Computer Organization EE-306 Computer Organization & Microprocessor VL-326 Computer Organization	—	E24 (p. 5)
CS-202 Data Structures and Algorithms (3-1-0-4)	—	IC-253 Programming and Data Structures	N25 (p. 6)
CS-205 Paradigms of Programming (3-0-0-3)	CS-302 Paradigms of Programming	—	E21 (p. 5)
CS-206 Computer Networks (3-0-0-3)	CS-313 Computer Networks CS-406 Computer Networks	—	E23 (p. 5)
CS-207 Applied Databases Practicum (0-0-3-2)	MA-323 Applied Databases Practicum	—	E12 (p. 5)
CS-208 Mathematical Foundations of Computer Science (3-1-0-4)	—	CS-513 Discrete Mathematics	N21 (p. 6)
CS-212 Design and Analysis of Algorithms (3-0-2-4)	MA-312 Design and Analysis of Algorithms	CS-403 Algorithm Design and Analysis CS-514 Data Structures and Algorithms-II CS-580 Advanced Data Structures and Algorithms (old)	E13 (p. 5) N14 (p. 5) N26 (p. 6)
CS-214 Computer Organization (3-0-2-4)	CS-201 Computer Organization EE-306 Computer Organization & Microprocessor VL-326 Computer Organization	—	E24 (p. 5)
CS-302 Paradigms of Programming (3-0-2-4)	CS-205 Paradigms of Programming	—	E21 (p. 5)
CS-303 Software Engineering (2-0-2-3)	—	CS-347 Software Engineering	N27 (p. 6)
CS-304 Formal Languages and Automata Theory (3-0-0-3)	MA-313 Formal Languages and Automata Theory	—	E14 (p. 5)
CS-305 Artificial Intelligence (3-0-0-3)	CS-315 Artificial Intelligence CS-362 Artificial Intelligence CS-562 Artificial Intelligence	—	E15 (p. 5)
CS-306 Operating Systems (3-0-0-3)	CS-312 Operating System	—	E20 (p. 5)
CS-312 Operating System (3-0-2-4)	CS-306 Operating Systems	—	E20 (p. 5)
CS-313 Computer Networks (3-0-2-4)	CS-206 Computer Networks CS-406 Computer Networks	—	E23 (p. 5)
CS-315 Artificial Intelligence (3-0-2-4)	CS-305 Artificial Intelligence CS-362 Artificial Intelligence CS-562 Artificial Intelligence	—	E15 (p. 5)
CS-347 Software Engineering (3-0-2-4)	—	CS-303 Software Engineering	N27 (p. 6)
CS-362 Artificial Intelligence (3-0-0-3)	CS-305 Artificial Intelligence CS-315 Artificial Intelligence CS-562 Artificial Intelligence	—	E15 (p. 5)
CS-403 Algorithm Design and Analysis (3-0-2-4)	—	CS-212 Design and Analysis of Algorithms CS-514 Data Structures and Algorithms-II CS-580 Advanced Data Structures and Algorithms (old) MA-312 Design and Analysis of Algorithms	N14 (p. 5) N23 (p. 6)
CS-406 Computer Networks (3-1-0-4)	CS-206 Computer Networks CS-313 Computer Networks	—	E23 (p. 5)
CS-501 Access Networks (3-0-0-3)	EE-533 Access Network	—	E25 (p. 5)
CS-508 Introduction to Heterogeneous Computing (2-0-0-2)	—	ME-522 High-Performance Scientific Computing	N89 (p. 15)

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Course (code, title, L-T-P-C)	Equivalent codes (only one may be credited)	Cannot be credited together with	Rows
CS-511 Applied Probability (2-0-0-2)	—	CS-513 Discrete Mathematics CS-601 Probability and Random Processes DS-303 Statistical Foundations of Data Science DS-313 Statistical Foundations of Data Science DS-314 Probability Theory and Statistics EE-534 Probability and Random Processes EE-581 Applied Statistics for Data and Signals IC-210 Probability, Statistics and Random Processes MA-524 Probability and Statistics	N15 (p. 5) N16 (p. 5) N17 (p. 5) N22 (p. 6) N33 (p. 7) N46 (p. 9)
CS-512 Matrix Theory (2-0-0-2)	—	EE-522 Matrix Theory for Engineers EE-553 Foundations of Intelligent Communication Systems-I IC-111 Linear Algebra IC-114 Linear Algebra MA-512 Linear Algebra	N18 (p. 5) N19 (p. 5) N20 (p. 5) N41 (p. 9)
CS-513 Discrete Mathematics (3-1-0-4)	—	CS-208 Mathematical Foundations of Computer Science CS-511 Applied Probability	N21 (p. 6) N22 (p. 6)
CS-514 Data Structures and Algorithms-II (3-0-2-4)	CS-580 Advanced Data Structures and Algorithms (old)	CS-212 Design and Analysis of Algorithms CS-403 Algorithm Design and Analysis MA-312 Design and Analysis of Algorithms	E16 (p. 5) N23 (p. 6) N26 (p. 6)
CS-520 Introduction to Quantum Computing (2-0-1-3)	QT-405 Introduction to Quantum Computation	PH-579 Quantum Computation and Information	E17 (p. 5) N24 (p. 6)
CS-562 Artificial Intelligence (3-0-0-3)	CS-305 Artificial Intelligence CS-315 Artificial Intelligence CS-362 Artificial Intelligence	—	E15 (p. 5)
CS-580 Advanced Data Structures and Algorithms (old) (3-0-1-4)	CS-514 Data Structures and Algorithms-II	CS-212 Design and Analysis of Algorithms CS-403 Algorithm Design and Analysis MA-312 Design and Analysis of Algorithms	E16 (p. 5) N23 (p. 6) N26 (p. 6)
CS-601 Probability and Random Processes (3-1-0-4)	EE-534 Probability and Random Processes	CS-511 Applied Probability DS-303 Statistical Foundations of Data Science DS-313 Statistical Foundations of Data Science DS-314 Probability Theory and Statistics EE-581 Applied Statistics for Data and Signals IC-210 Probability, Statistics and Random Processes MA-524 Probability and Statistics	E38 (p. 8) N15 (p. 5) N34 (p. 8) N47 (p. 9) N57 (p. 10) N60 (p. 10)
CS-603 Managerial Decision Making (3-0-0-3)	HS-616 Managerial Thinking and Decision Making	—	E22 (p. 5)
CS-609 Speech Processing (3-0-2-4)	—	CS-630 Speech Technology	N28 (p. 6)
CS-630 Speech Technology (3-0-2-4)	—	CS-609 Speech Processing	N28 (p. 6)
CS-669 Pattern Recognition (3-1-0-4)	DS-403 Introduction to Statistical Learning DS-413 Introduction to Statistical Learning DS-416 Fundamentals of Machine Learning	MA-574 An Introduction to Probabilistic Machine Learning	E35 (p. 7) N82 (p. 14)
CS-671 Deep Learning and Applications (3-1-0-4)	DS-417 Deep Learning	—	E18 (p. 5)
CS-683 Generative AI (3-0-1-4)	DS-418 Introduction to Generative AI	—	E19 (p. 5)
CY-001 Preparatory Chemistry - 1 (3-1-0-4)	CY-100 Preparatory Chemistry 1	—	E26 (p. 6)
CY-100 Preparatory Chemistry 1 (3-0-0-3)	CY-001 Preparatory Chemistry - 1	—	E26 (p. 6)
CY-241 Nano-scale Science and Technology (3-0-0-3)	—	CY-554 Science and Technology of Nanomaterials	N29 (p. 7)
CY-501P Organic Chemistry Laboratory (0-0-8-4)	CY-531P Organic Chemistry Laboratory	—	E27 (p. 6)
CY-502 Photochemistry and pericyclic reactions (3-1-0-4)	CY-532 Photochemistry and Pericyclic Reactions	—	E28 (p. 6)

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Course (code, title, L-T-P-C)	Equivalent codes (only one may be credited)	Cannot be credited together with	Rows
CY-506 Chemistry of Transition Elements (3-1-0-4)	CY-534 Chemistry of Transition Elements	—	E31 (p. 6)
CY-506P Inorganic Chemistry Lab (0-0-8-4)	CY-533P Inorganic Chemistry Lab	—	E30 (p. 6)
CY-507 Chemistry of Main Group Elements (3-1-0-4)	CY-533 Chemistry of Main Group Elements	—	E29 (p. 6)
CY-511 Group Theory and Spectroscopy (3-0-0-3)	—	CY-701 Advances Physical Methods in Chemistry Theory and Applications	N32 (p. 7)
CY-515 Advanced Inorganic Spectroscopy (3-0-0-3)	—	CY-553 Organic Inorganic Spectroscopy	N30 (p. 7)
CY-523 Colloids and Interface Science and Technology (3-0-0-3)	—	MT-510 Colloids and Interfaces	N31 (p. 7)
CY-531P Organic Chemistry Laboratory (3 cr.)	CY-501P Organic Chemistry Laboratory	—	E27 (p. 6)
CY-532 Photochemistry and Pericyclic Reactions (3 cr.)	CY-502 Photochemistry and pericyclic reactions	—	E28 (p. 6)
CY-533 Chemistry of Main Group Elements (3 cr.)	CY-507 Chemistry of Main Group Elements	—	E29 (p. 6)
CY-533P Inorganic Chemistry Lab (0-0-8-4)	CY-506P Inorganic Chemistry Lab	—	E30 (p. 6)
CY-534 Chemistry of Transition Elements (3-1-0-4)	CY-506 Chemistry of Transition Elements	—	E31 (p. 6)
CY-552 Hydrogen Generation and Storage (3-0-0-3)	—	MT-509 Hydrogen Energy	N97 (p. 16)
CY-553 Organic Inorganic Spectroscopy (3-0-0-3)	—	CY-515 Advanced Inorganic Spectroscopy	N30 (p. 7)
CY-554 Science and Technology of Nanomaterials (3-0-0-3)	—	CY-241 Nano-scale Science and Technology	N29 (p. 7)
CY-600 Research Methodology for Chemistry (1-0-0-1)	BY-600 Research Methodology CE-600 Research Methodology for Civil Engineering HC-600 Research Methodology HS-600 Research Methodology IK-509 Research methodology MA-600 Research Methodology ME-600 Research Methodology PH-600 Research Methodology RM-510 Research Methodology RM-600 Research Methodology SC-600 Research Methodology	—	E99 (p. 17)
CY-701 Advances Physical Methods in Chemistry Theory and Applications (3-0-0-3)	—	CY-511 Group Theory and Spectroscopy	N32 (p. 7)
DS-302 Computing Systems for Data Processing (2-0-2-3)	DS-312 Computing Systems for Data Processing MA-120 Introduction to Computing System and Databases	—	E33 (p. 7)
DS-303 Statistical Foundations of Data Science (3-0-0-3)	DS-313 Statistical Foundations of Data Science DS-314 Probability Theory and Statistics	CS-511 Applied Probability CS-601 Probability and Random Processes EE-534 Probability and Random Processes IC-210 Probability, Statistics and Random Processes MA-524 Probability and Statistics MB-510 Probability and Statistics for Data Science and AI	E32 (p. 7) N33 (p. 7) N34 (p. 8) N35 (p. 8) N36 (p. 8) N38 (p. 8)
DS-312 Computing Systems for Data Processing (3-0-2-4)	DS-302 Computing Systems for Data Processing MA-120 Introduction to Computing System and Databases	—	E33 (p. 7)

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Course (code, title, L-T-P-C)	Equivalent codes (only one may be credited)	Cannot be credited together with	Rows
DS-313 Statistical Foundations of Data Science (3-1-0-4)	DS-303 Statistical Foundations of Data Science DS-314 Probability Theory and Statistics	CS-511 Applied Probability CS-601 Probability and Random Processes EE-534 Probability and Random Processes IC-210 Probability, Statistics and Random Processes MA-524 Probability and Statistics MB-510 Probability and Statistics for Data Science and AI	E32 (p. 7) N33 (p. 7) N34 (p. 8) N35 (p. 8) N36 (p. 8) N38 (p. 8)
DS-314 Probability Theory and Statistics (3-1-0-4)	DS-303 Statistical Foundations of Data Science DS-313 Statistical Foundations of Data Science	CS-511 Applied Probability CS-601 Probability and Random Processes EE-534 Probability and Random Processes IC-210 Probability, Statistics and Random Processes MA-524 Probability and Statistics MB-510 Probability and Statistics for Data Science and AI	E32 (p. 7) N33 (p. 7) N34 (p. 8) N35 (p. 8) N36 (p. 8) N38 (p. 8)
DS-401 Optimization for Data Science (3-0-0-3)	DS-411 Optimization for Data Science DS-415 Optimization for Machine Learning	EE-530 Applied Optimization MA-576 Nonlinear Optimization	E34 (p. 7) N37 (p. 8) N39 (p. 8)
DS-402 Matrix Computations for Data Science (2-0-2-3)	DS-412 Matrix Computations for Data Science	MA-310 Matrix Computations & Lab	E36 (p. 7) N79 (p. 13)
DS-403 Introduction to Statistical Learning (2-0-2-3)	CS-669 Pattern Recognition DS-413 Introduction to Statistical Learning DS-416 Fundamentals of Machine Learning	MA-574 An Introduction to Probabilistic Machine Learning	E35 (p. 7) N82 (p. 14)
DS-411 Optimization for Data Science (3-1-0-4)	DS-401 Optimization for Data Science DS-415 Optimization for Machine Learning	EE-530 Applied Optimization MA-576 Nonlinear Optimization	E34 (p. 7) N37 (p. 8) N39 (p. 8)
DS-412 Matrix Computations for Data Science (3-0-2-4)	DS-402 Matrix Computations for Data Science	MA-310 Matrix Computations & Lab	E36 (p. 7) N79 (p. 13)
DS-413 Introduction to Statistical Learning (3-1-0-4)	CS-669 Pattern Recognition DS-403 Introduction to Statistical Learning DS-416 Fundamentals of Machine Learning	MA-574 An Introduction to Probabilistic Machine Learning	E35 (p. 7) N82 (p. 14)
DS-415 Optimization for Machine Learning (3-0-2-4)	DS-401 Optimization for Data Science DS-411 Optimization for Data Science	EE-530 Applied Optimization MA-576 Nonlinear Optimization	E34 (p. 7) N37 (p. 8) N39 (p. 8)
DS-416 Fundamentals of Machine Learning (3-0-2-4)	CS-669 Pattern Recognition DS-403 Introduction to Statistical Learning DS-413 Introduction to Statistical Learning	MA-574 An Introduction to Probabilistic Machine Learning	E35 (p. 7) N82 (p. 14)
DS-417 Deep Learning (3-0-2-4)	CS-671 Deep Learning and Applications	—	E18 (p. 5)
DS-418 Introduction to Generative AI (3-0-2-4)	CS-683 Generative AI	—	E19 (p. 5)
EE-101 Electrical Science (3-1-2-4)	EE-261 Electrical Systems Around Us IC-160 Electrical Systems Around Us	—	E39 (p. 8)
EE-205 Electromagnetics and Wave Propagation (3-0-0-3)	—	PH-421 Electromagnetic Theory PH-521 Electromagnetic Theory	N40 (p. 9)
EE-210 Digital System Design (3-0-0-3)	EE-212 Digital System Design	—	E45 (p. 8)
EE-211 Analog Circuit Design (2-0-2-3)	—	EE-677 Analog Circuit Design	N58 (p. 10)
EE-212 Digital System Design (3-0-2-4)	EE-210 Digital System Design	—	E45 (p. 8)
EE-231 Measurement and Instrumentation (2-0-2-3)	IC-231 Measurement and Instrumentation	EE-313 Measurement and Instrumentation	E42 (p. 8) N61 (p. 10)
EE-260 Signals and Systems (2.5-0.5-0-3)	IC-260 Signals and Systems	—	E46 (p. 8)
EE-261 Electrical Systems Around Us (3-0-0-3)	EE-101 Electrical Science IC-160 Electrical Systems Around Us	—	E39 (p. 8)
EE-301 Control Systems (old version) (2.5-0.5-0-3)	EE-302 Control Systems EE-602 Control System Applications	—	E44 (p. 8)
EE-302 Control Systems (3-0-2-4)	EE-301 Control Systems (old version) EE-602 Control System Applications	—	E44 (p. 8)
EE-303 Power Systems (3-1-0-4)	—	EE-552 Power and Energy Systems	N55 (p. 9)
EE-304 Communication Systems (3-0-1-4)	EE-316 Communication Systems	—	E47 (p. 8)
EE-305 Digital Signal Processing (3-1-0-4)	EE-314 Digital Signal Processing	—	E37 (p. 8)

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Course (code, title, L-T-P-C)	Equivalent codes (only one may be credited)	Cannot be credited together with	Rows
EE-306 Computer Organization & Microprocessor (3-0-0-3)	CS-201 Computer Organization CS-214 Computer Organization VL-326 Computer Organization	—	E24 (p. 5)
EE-311 Device Electronics for Integrated Circuits (3-0-0-3)	—	EE-520 Microelectronics Devices and Modelling	N59 (p. 10)
EE-313 Measurement and Instrumentation (2-1-0-3)	—	EE-231 Measurement and Instrumentation IC-231 Measurement and Instrumentation	N61 (p. 10)
EE-314 Digital Signal Processing (3-0-2-4)	EE-305 Digital Signal Processing	—	E37 (p. 8)
EE-316 Communication Systems (3-0-2-4)	EE-304 Communication Systems	—	E47 (p. 8)
EE-503 Advance Communication Theory (3-0-0-3)	EE-570 Advanced Communication Theory	—	E43 (p. 8)
EE-508 Fundamentals of Electric Drives (3-0-0-3)	—	EE-601 Advanced Electric Drives EE-604 Advanced Electric Drives	N62 (p. 10)
EE-512 CMOS Analog IC Design (3-0-2-4)	VL-404 CMOS Analog IC Design	—	E48 (p. 9)
EE-518 Information Theory (3-0-0-3)	EE-605 Information Theory	—	E40 (p. 8)
EE-520 Microelectronics Devices and Modelling (3-0-0-3)	—	EE-311 Device Electronics for Integrated Circuits	N59 (p. 10)
EE-522 Matrix Theory for Engineers (3-0-0-3)	—	CS-512 Matrix Theory EE-553 Foundations of Intelligent Communication Systems-I	N18 (p. 5) N42 (p. 9)
EE-524 Digital MOS LSI Circuits (3-0-0-3)	—	EE-678 Digital Circuit Design	N54 (p. 9)
EE-530 Applied Optimization (3-0-0-3)	—	DS-401 Optimization for Data Science DS-411 Optimization for Data Science DS-415 Optimization for Machine Learning	N37 (p. 8)
EE-531 Estimation and Detection Theory (3-0-0-3)	—	EE-575 Applied Statistics for Data and Signals	N56 (p. 9)
EE-533 Access Network (4 cr.)	CS-501 Access Networks	—	E25 (p. 5)
EE-534 Probability and Random Processes (3-0-0-3)	CS-601 Probability and Random Processes	CS-511 Applied Probability DS-303 Statistical Foundations of Data Science DS-313 Statistical Foundations of Data Science DS-314 Probability Theory and Statistics EE-581 Applied Statistics for Data and Signals IC-210 Probability, Statistics and Random Processes MA-524 Probability and Statistics	E38 (p. 8) N15 (p. 5) N34 (p. 8) N47 (p. 9) N57 (p. 10) N60 (p. 10)
EE-543 Vision and Learning Based Control (3-0-0-3)	AR-514 Vision and Learning Based Control	—	E3 (p. 3)
EE-552 Power and Energy Systems (3-1-0-4)	—	EE-303 Power Systems	N55 (p. 9)
EE-553 Foundations of Intelligent Communication Systems-I (3-0-2-4)	—	CS-512 Matrix Theory EE-522 Matrix Theory for Engineers MA-221 Numerical Analysis MA-512 Linear Algebra MA-523 Numerical Analysis MA-525 Heuristic Optimization MA-551 Numerical Analysis MA-607 Numerical Analysis	N41 (p. 9) N42 (p. 9) N43 (p. 9) N44 (p. 9) N45 (p. 9)
EE-555 Intelligent Control System (3-0-0-3)	EE-597 Intelligent Control System	—	E49 (p. 9)
EE-570 Advanced Communication Theory (3-0-2-4)	EE-503 Advance Communication Theory	—	E43 (p. 8)
EE-575 Applied Statistics for Data and Signals (3-0-2-4)	—	EE-531 Estimation and Detection Theory EE-581 Applied Statistics for Data and Signals	N56 (p. 9) N63 (p. 10)

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Course (code, title, L-T-P-C)	Equivalent codes (only one may be credited)	Cannot be credited together with	Rows
EE-581 Applied Statistics for Data and Signals (3-0-2-4)	—	CS-511 Applied Probability CS-601 Probability and Random Processes EE-534 Probability and Random Processes EE-575 Applied Statistics for Data and Signals MA-511 Real Analysis MA-521 Functional Analysis MA-524 Probability and Statistics MA-568 Real Analysis MA-601 Real and Functional Analysis	N46 (p. 9) N47 (p. 9) N48 (p. 9) N49 (p. 9) N50 (p. 9) N51 (p. 9) N52 (p. 9) N63 (p. 10)
EE-597 Intelligent Control System (3-0-0-3)	EE-555 Intelligent Control System	—	E49 (p. 9)
EE-601 Advanced Electric Drives (2.5-0.5-0-3)	EE-604 Advanced Electric Drives	EE-508 Fundamentals of Electric Drives	E50 (p. 9) N62 (p. 10)
EE-602 Control System Applications (2.5-0.5-0-3)	EE-301 Control Systems (old version) EE-302 Control Systems	—	E44 (p. 8)
EE-604 Advanced Electric Drives (3 cr.)	EE-601 Advanced Electric Drives	EE-508 Fundamentals of Electric Drives	E50 (p. 9) N62 (p. 10)
EE-605 Information Theory (3 cr.)	EE-518 Information Theory	—	E40 (p. 8)
EE-607 Optical Communication Systems (3-0-0-3)	EE-614 Optical Communication Systems	—	E41 (p. 8)
EE-614 Optical Communication Systems (3-0-0-3)	EE-607 Optical Communication Systems	—	E41 (p. 8)
EE-616 Microwave engineering (2-0-0-2)	—	EE-622 Microwave Integrated Circuits	N53 (p. 9)
EE-622 Microwave Integrated Circuits (3-0-0-3)	—	EE-616 Microwave engineering	N53 (p. 9)
EE-677 Analog Circuit Design (3 cr.)	—	EE-211 Analog Circuit Design	N58 (p. 10)
EE-678 Digital Circuit Design (3 cr.)	—	EE-524 Digital MOS LSI Circuits	N54 (p. 9)
EM-504 Materials for Energy Applications (3-1-0-4)	EM-601 (title not recorded)	—	E51 (p. 10)
EM-601 (title not recorded) (L-T-P-C not recorded)	EM-504 Materials for Energy Applications	—	E51 (p. 10)
EM-604 Energy: Environment Policy and Law (3-1-0-4)	EN-504 Energy: Environment Policy and Law EN-604 Energy: Environment Policy and Law	—	E54 (p. 10)
EM-658 Deformation Behavior of Energy Materials (3-0-0-3)	EN-611 Durability Behavior of Energy Materials	MT-304 Mechanical Behavior of Materials	E53 (p. 10) N65 (p. 11)
EN-503 Energy Storage Technologies (3-0-0-3)	—	MT-501 Energy Conversion & Storage Technology	N96 (p. 16)
EN-504 Energy: Environment Policy and Law (3-0-0-3)	EM-604 Energy: Environment Policy and Law EN-604 Energy: Environment Policy and Law	—	E54 (p. 10)
EN-506 Design of Energy Systems (3-0-0-3)	ME-633 Design of Energy Systems	—	E91 (p. 15)
EN-509 Functional Materials for Energy Engineering (3-0-0-3)	ME-502 (old) Functional Materials (old ME-502) ME-609 Functional Materials MT-205 Functional Properties of Materials	—	E52 (p. 10)
EN-509 (old) Thermodynamics for Energy Systems (old EN-509) (3-0-0-3)	ME-634 Thermodynamics for Energy Systems	IC-142 Engineering Thermodynamics ME-213 Engineering Thermodynamics ME-610 Advanced Thermodynamics	E85 (p. 15) N90 (p. 15) N93 (p. 15)
EN-512 Structure-Property Correlations for Energy Applications (1-0-4-3)	—	EN-612 Structure - Property correlation in materials for Energy Applications MT-203 Materials Synthesis & Characterization	N66 (p. 11) N95 (p. 16)
EN-604 Energy: Environment Policy and Law (3-0-0-3)	EM-604 Energy: Environment Policy and Law EN-504 Energy: Environment Policy and Law	—	E54 (p. 10)

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Course (code, title, L-T-P-C)	Equivalent codes (only one may be credited)	Cannot be credited together with	Rows
EN-605 Modeling and Simulation of Energy Systems (2-0-2-3)	—	ME-620 Modeling and Simulation	N64 (p. 11)
EN-611 Durability Behavior of Energy Materials (3-0-0-3)	EM-658 Deformation Behavior of Energy Materials	MT-304 Mechanical Behavior of Materials	E53 (p. 10) N65 (p. 11)
EN-612 Structure - Property correlation in materials for Energy Applications (2-0-2-3)	—	EN-512 Structure-Property Correlations for Energy Applications	N66 (p. 11)
EP-301 Engineering Mathematics-2 (3-1-0-4)	—	PH-511 Mathematical Physics	N100 (p. 17)
EP-321 Foundations of Electrodynamics (3-0-0-3)	IC-221 Foundations of Electrodynamics	—	E98 (p. 16)
EP-403 Physics of atoms and molecules (3-0-0-3)	—	PH-424 Atomic and Molecular Spectroscopy PH-524 Atomic and Molecular Physics	N98 (p. 16) N103 (p. 17)
GE-521 Essentials of Entrepreneurship (3-0-0-3)	HS-510 Essentials of Entrepreneurship	—	E55 (p. 11)
HC-600 Research Methodology (1-0-0-1)	BY-600 Research Methodology CE-600 Research Methodology for Civil Engineering CY-600 Research Methodology for Chemistry HS-600 Research Methodology IK-509 Research methodology MA-600 Research Methodology ME-600 Research Methodology PH-600 Research Methodology RM-510 Research Methodology RM-600 Research Methodology SC-600 Research Methodology	—	E99 (p. 17)
HS-205 Financial Accounting (3-0-0-3)	MB-303 Financial Accounting	—	E63 (p. 11)
HS-253 Introduction to Sociology (3-0-0-3)	HS-344 Introduction to Sociology	—	E60 (p. 11)
HS-304 Organizational Management (3-0-0-3)	—	MB-201 Foundations of Business Management	N88 (p. 14)
HS-305 Science Writing (3-0-0-3)	HS-358 Science Writing	—	E57 (p. 11)
HS-307 Macroeconomics I (3-0-0-3)	MB-203 Macroeconomics	—	E56 (p. 11)
HS-344 Introduction to Sociology (3-0-0-3)	HS-253 Introduction to Sociology	—	E60 (p. 11)
HS-358 Science Writing (3-0-0-3)	HS-305 Science Writing	—	E57 (p. 11)
HS-364 Modern China: A Historical Survey (3-0-0-3)	HS-392 Modern China	—	E58 (p. 11)
HS-382 Social Movements in India (3-0-0-3)	—	HS-536 Social Movements in India	N68 (p. 12)
HS-392 Modern China (3-0-0-3)	HS-364 Modern China: A Historical Survey	—	E58 (p. 11)
HS-403 Organizational Behavior (3-0-0-3)	—	MB-524 Organizational Behaviour	N70 (p. 12)
HS-450 Financial Management (3-0-0-3)	HS-551 Financial Management	MB-527 Financial Management	E61 (p. 11) N69 (p. 12)
HS-461 Consumer Behavior (3-0-0-3)	—	MB-582 Consumer Behavior	N67 (p. 12)
HS-471 Contemporary India (3-0-0-3)	HS-537 Post-Reform India: Polity, Society and Economy	—	E59 (p. 11)
HS-510 Essentials of Entrepreneurship (3-0-0-3)	GE-521 Essentials of Entrepreneurship	—	E55 (p. 11)
HS-536 Social Movements in India (3-0-0-3)	—	HS-382 Social Movements in India	N68 (p. 12)
HS-537 Post-Reform India: Polity, Society and Economy (3-0-0-3)	HS-471 Contemporary India	—	E59 (p. 11)
HS-550 Statistical Methods (3-0-2-4)	HS-650 Statistical Methods	MA-611 Statistical tools and Computing	E62 (p. 11) N83 (p. 14)
HS-551 Financial Management (3-0-0-3)	HS-450 Financial Management	MB-527 Financial Management	E61 (p. 11) N69 (p. 12)

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Course (code, title, L-T-P-C)	Equivalent codes (only one may be credited)	Cannot be credited together with	Rows
HS-600 Research Methodology (1-0-0-1)	BY-600 Research Methodology CE-600 Research Methodology for Civil Engineering CY-600 Research Methodology for Chemistry HC-600 Research Methodology IK-509 Research methodology MA-600 Research Methodology ME-600 Research Methodology PH-600 Research Methodology RM-510 Research Methodology RM-600 Research Methodology SC-600 Research Methodology	—	E99 (p. 17)
HS-616 Managerial Thinking and Decision Making (3-0-0-3)	CS-603 Managerial Decision Making	—	E22 (p. 5)
HS-650 Statistical Methods (3-0-0-3)	HS-550 Statistical Methods	MA-611 Statistical tools and Computing	E62 (p. 11) N83 (p. 14)
IC-110 Engineering Mathematics (2.5-0.5-0-3)	—	IC-112 Calculus MA-101 Mathematics-I	N71 (p. 12) N73 (p. 12)
IC-111 Linear Algebra (2.5-0.5-0-3)	IC-114 Linear Algebra	CS-512 Matrix Theory MA-512 Linear Algebra	E66 (p. 12) N19 (p. 5) N75 (p. 12)
IC-112 Calculus (1.5-0.5-0-2)	—	IC-110 Engineering Mathematics MA-101 Mathematics-I	N71 (p. 12) N74 (p. 12)
IC-114 Linear Algebra (2-0-0-2)	IC-111 Linear Algebra	CS-512 Matrix Theory MA-512 Linear Algebra	E66 (p. 12) N19 (p. 5) N75 (p. 12)
IC-130 Applied Chemistry for Engineers (3-0-0-3)	—	IC-131 Applied Chemistry for Engineers	N77 (p. 13)
IC-131 Applied Chemistry for Engineers (2-0-2-3)	—	IC-130 Applied Chemistry for Engineers	N77 (p. 13)
IC-140 Engineering Graphics for Design (2-0-3-4)	CE-101 Engineering Graphics	—	E64 (p. 12)
IC-141 Product Realization Technology (2-0-0-2)	ME-212 Product Manufacturing Technology	—	E89 (p. 15)
IC-142 Engineering Thermodynamics (2.5-0.5-0-3)	ME-213 Engineering Thermodynamics	EN-509 (old) Thermodynamics for Energy Systems (old EN-509) ME-634 Thermodynamics for Energy Systems MI-101 Thermodynamics	E90 (p. 15) N91 (p. 15) N93 (p. 15)
IC-160 Electrical Systems Around Us (2.5-0.5-0-3)	EE-101 Electrical Science EE-261 Electrical Systems Around Us	—	E39 (p. 8)
IC-210 Probability, Statistics and Random Processes (2.5-0.5-0-3)	—	CS-511 Applied Probability CS-601 Probability and Random Processes DS-303 Statistical Foundations of Data Science DS-313 Statistical Foundations of Data Science DS-314 Probability Theory and Statistics EE-534 Probability and Random Processes MA-524 Probability and Statistics	N16 (p. 5) N38 (p. 8) N57 (p. 10) N76 (p. 12)
IC-221 Foundations of Electrodynamics (2.5-0.5-0-3)	EP-321 Foundations of Electrodynamics	—	E98 (p. 16)
IC-230 Environmental Science (3-0-0-3)	CE-102 Environmental Studies	—	E65 (p. 12)
IC-231 Measurement and Instrumentation (2-0-2-3)	EE-231 Measurement and Instrumentation	EE-313 Measurement and Instrumentation	E42 (p. 8) N61 (p. 10)
IC-241 Materials Science for Engineers (3-0-0-3)	ME-204 Materials Science for Engineers	—	E67 (p. 12)
IC-242 Continuum Mechanics (2.5-0.5-0-3)	—	CE-530 Continuum Mechanics ME-530 Continuum Mechanics	N72 (p. 12)
IC-253 Programming and Data Structures (2-0-2-3)	—	CS-202 Data Structures and Algorithms	N25 (p. 6)
IC-260 Signals and Systems (2.5-0.5-0-3)	EE-260 Signals and Systems	—	E46 (p. 8)

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Course (code, title, L-T-P-C)	Equivalent codes (only one may be credited)	Cannot be credited together with	Rows
IK-101 Sanskrit Language – Level I (UG) (L-T-P-C not recorded)	IK-527 Sanskrit Language – Level I (PG)	—	E68 (p. 13)
IK-504 Bhagavad Gita Comprehensive (3-0-0-3)	IK-530 Bhagavad-Gītā Part I	—	E69 (p. 13)
IK-509 Research methodology (1-0-0-1)	BY-600 Research Methodology CE-600 Research Methodology for Civil Engineering CY-600 Research Methodology for Chemistry HC-600 Research Methodology HS-600 Research Methodology MA-600 Research Methodology ME-600 Research Methodology PH-600 Research Methodology RM-510 Research Methodology RM-600 Research Methodology SC-600 Research Methodology	—	E99 (p. 17)
IK-517 Philosophy of Science (3-0-0-3)	IK-593 Philosophy of Science	—	E70 (p. 13)
IK-527 Sanskrit Language – Level I (PG) (3-0-0-3)	IK-101 Sanskrit Language – Level I (UG)	—	E68 (p. 13)
IK-530 Bhagavad-Gītā Part I (3-0-0-3)	IK-504 Bhagavad Gita Comprehensive	—	E69 (p. 13)
IK-593 Philosophy of Science (3 cr.)	IK-517 Philosophy of Science	—	E70 (p. 13)
MA-101 Mathematics-I (3-1-0-4)	—	IC-110 Engineering Mathematics IC-112 Calculus	N73 (p. 12) N74 (p. 12)
MA-120 Introduction to Computing System and Databases (3-0-2-4)	DS-302 Computing Systems for Data Processing DS-312 Computing Systems for Data Processing	—	E33 (p. 7)
MA-210 Real and Complex Analysis (2.5-0.5-0-3)	—	MA-511 Real Analysis	N84 (p. 14)
MA-211 Ordinary Differential Equations (3-1-0-4)	MA-513 Ordinary Differential Equations	MA-465 Ordinary Differential Equations	E71 (p. 13) N86 (p. 14)
MA-220 Partial Differential Equations (3-1-0-4)	MA-522 Partial Differential Equations	—	E72 (p. 13)
MA-221 Numerical Analysis (3-1-0-4)	MA-523 Numerical Analysis MA-551 Numerical Analysis MA-607 Numerical Analysis	EE-553 Foundations of Intelligent Communication Systems-I	E75 (p. 13) N43 (p. 9)
MA-222 Applied Linear programming (3-1-0-4)	—	MA-515 Applied Mathematical Programming	N78 (p. 13)
MA-251 Abstract Algebra (3-0-0-3)	MA-548 Abstract Algebra MA-549 Abstract Algebra	—	E74 (p. 13)
MA-310 Matrix Computations & Lab (3-0-2-4)	—	DS-402 Matrix Computations for Data Science DS-412 Matrix Computations for Data Science	N79 (p. 13)
MA-312 Design and Analysis of Algorithms (3-0-2-4)	CS-212 Design and Analysis of Algorithms	CS-403 Algorithm Design and Analysis CS-514 Data Structures and Algorithms-II CS-580 Advanced Data Structures and Algorithms (old)	E13 (p. 5) N14 (p. 5) N26 (p. 6)
MA-313 Formal Languages and Automata Theory (3-0-0-3)	CS-304 Formal Languages and Automata Theory	—	E14 (p. 5)
MA-321 Numerics of Differential Equations (3-0-2-4)	—	MA-609 Numerics of Partial Differential Equations	N80 (p. 14)
MA-322 Applied Graph Theory (3-0-2-4)	—	MA-530 Graph Theory	N81 (p. 14)
MA-323 Applied Databases Practicum (0-0-3-2)	CS-207 Applied Databases Practicum	—	E12 (p. 5)
MA-460 Nonlinear Dynamics and Chaos (3-0-0-3)	MA-560 Nonlinear Dynamics and Chaos	—	E73 (p. 13)
MA-465 Ordinary Differential Equations (3-0-0-3)	—	MA-211 Ordinary Differential Equations MA-513 Ordinary Differential Equations	N86 (p. 14)

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Course (code, title, L-T-P-C)	Equivalent codes (only one may be credited)	Cannot be credited together with	Rows
MA-511 Real Analysis (3-1-0-4)	—	EE-581 Applied Statistics for Data and Signals MA-210 Real and Complex Analysis MA-568 Real Analysis	N48 (p. 9) N84 (p. 14) N85 (p. 14)
MA-512 Linear Algebra (3-1-0-4)	—	CS-512 Matrix Theory EE-553 Foundations of Intelligent Communication Systems-I IC-111 Linear Algebra IC-114 Linear Algebra	N20 (p. 5) N44 (p. 9) N75 (p. 12)
MA-513 Ordinary Differential Equations (3-1-0-4)	MA-211 Ordinary Differential Equations	MA-465 Ordinary Differential Equations	E71 (p. 13) N86 (p. 14)
MA-515 Applied Mathematical Programming (3-1-0-4)	—	MA-222 Applied Linear programming	N78 (p. 13)
MA-521 Functional Analysis (3-1-0-4)	—	EE-581 Applied Statistics for Data and Signals MA-601 Real and Functional Analysis	N49 (p. 9) N87 (p. 14)
MA-522 Partial Differential Equations (3-1-0-4)	MA-220 Partial Differential Equations	—	E72 (p. 13)
MA-523 Numerical Analysis (3-1-0-4)	MA-221 Numerical Analysis MA-551 Numerical Analysis MA-607 Numerical Analysis	EE-553 Foundations of Intelligent Communication Systems-I	E75 (p. 13) N43 (p. 9)
MA-524 Probability and Statistics (3-1-0-4)	—	CS-511 Applied Probability CS-601 Probability and Random Processes DS-303 Statistical Foundations of Data Science DS-313 Statistical Foundations of Data Science DS-314 Probability Theory and Statistics EE-534 Probability and Random Processes EE-581 Applied Statistics for Data and Signals IC-210 Probability, Statistics and Random Processes	N17 (p. 5) N35 (p. 8) N50 (p. 9) N60 (p. 10) N76 (p. 12)
MA-525 Heuristic Optimization (3-0-0-3)	—	EE-553 Foundations of Intelligent Communication Systems-I	N45 (p. 9)
MA-530 Graph Theory (3-1-0-4)	—	MA-322 Applied Graph Theory	N81 (p. 14)
MA-548 Abstract Algebra (3-1-0-4)	MA-251 Abstract Algebra MA-549 Abstract Algebra	—	E74 (p. 13)
MA-549 Abstract Algebra (3-0-0-3)	MA-251 Abstract Algebra MA-548 Abstract Algebra	—	E74 (p. 13)
MA-550 Statistical Data Analysis (2-1-0-3)	MA-605 Statistical Data Analysis	—	E76 (p. 13)
MA-551 Numerical Analysis (3-0-0-3)	MA-221 Numerical Analysis MA-523 Numerical Analysis MA-607 Numerical Analysis	EE-553 Foundations of Intelligent Communication Systems-I	E75 (p. 13) N43 (p. 9)
MA-560 Nonlinear Dynamics and Chaos (3-0-0-3)	MA-460 Nonlinear Dynamics and Chaos	—	E73 (p. 13)
MA-568 Real Analysis (2.5-0.5-0-3)	—	EE-581 Applied Statistics for Data and Signals MA-511 Real Analysis	N51 (p. 9) N85 (p. 14)
MA-574 An Introduction to Probabilistic Machine Learning (3-0-2-4)	—	CS-669 Pattern Recognition DS-403 Introduction to Statistical Learning DS-413 Introduction to Statistical Learning DS-416 Fundamentals of Machine Learning	N82 (p. 14)
MA-576 Nonlinear Optimization (3-0-2-4)	—	DS-401 Optimization for Data Science DS-411 Optimization for Data Science DS-415 Optimization for Machine Learning	N39 (p. 8)

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Course (code, title, L-T-P-C)	Equivalent codes (only one may be credited)	Cannot be credited together with	Rows
MA-600 Research Methodology (1-0-0-1)	BY-600 Research Methodology CE-600 Research Methodology for Civil Engineering CY-600 Research Methodology for Chemistry HC-600 Research Methodology HS-600 Research Methodology IK-509 Research methodology ME-600 Research Methodology PH-600 Research Methodology RM-510 Research Methodology RM-600 Research Methodology SC-600 Research Methodology	—	E99 (p. 17)
MA-601 Real and Functional Analysis (2-1-0-3)	—	EE-581 Applied Statistics for Data and Signals MA-521 Functional Analysis	N52 (p. 9) N87 (p. 14)
MA-605 Statistical Data Analysis (2-1-0-3)	MA-550 Statistical Data Analysis	—	E76 (p. 13)
MA-607 Numerical Analysis (2-0-2-3)	MA-221 Numerical Analysis MA-523 Numerical Analysis MA-551 Numerical Analysis	EE-553 Foundations of Intelligent Communication Systems-I	E75 (p. 13) N43 (p. 9)
MA-609 Numerics of Partial Differential Equations (3-0-0-3)	—	MA-321 Numerics of Differential Equations	N80 (p. 14)
MA-611 Statistical tools and Computing (3-1-0-4)	—	HS-550 Statistical Methods HS-650 Statistical Methods	N83 (p. 14)
MA-641 Operations Research (3-0-0-3)	MB-402 Operations Research	—	E78 (p. 13)
MA-665 Semigroup Of Bounded Linear Operators (3 cr.)	MA-780 Topics on Semigroup Theory	—	E77 (p. 13)
MA-780 Topics on Semigroup Theory (3-0-0-3)	MA-665 Semigroup Of Bounded Linear Operators	—	E77 (p. 13)
MB-201 Foundations of Business Management (3-0-1-4)	—	HS-304 Organizational Management	N88 (p. 14)
MB-203 Macroeconomics (3 cr.)	HS-307 Macroeconomics I	—	E56 (p. 11)
MB-303 Financial Accounting (3 cr.)	HS-205 Financial Accounting	—	E63 (p. 11)
MB-402 Operations Research (3 cr.)	MA-641 Operations Research	—	E78 (p. 13)
MB-510 Probability and Statistics for Data Science and AI (2-0-0-2)	—	DS-303 Statistical Foundations of Data Science DS-313 Statistical Foundations of Data Science DS-314 Probability Theory and Statistics	N36 (p. 8)
MB-524 Organizational Behaviour (2-0-0-2)	—	HS-403 Organizational Behavior	N70 (p. 12)
MB-527 Financial Management (2-0-0-2)	—	HS-450 Financial Management HS-551 Financial Management	N69 (p. 12)
MB-554 Blockchain for Business (2-0-0-2)	—	AR-516 Introduction to Blockchain and Web3	N1 (p. 3)
MB-582 Consumer Behavior (2-0-0-2)	—	HS-461 Consumer Behavior	N67 (p. 12)
ME-204 Materials Science for Engineers (3-0-0-3)	IC-241 Materials Science for Engineers	—	E67 (p. 12)
ME-210 Fluid Mechanics (2.5-0.5-0-3)	CE-251 Hydraulics Engineering	—	E10 (p. 4)
ME-212 Product Manufacturing Technology (2-0-3-3)	IC-141 Product Realization Technology	—	E89 (p. 15)
ME-213 Engineering Thermodynamics (3-1-0-4)	IC-142 Engineering Thermodynamics	EN-509 (old) Thermodynamics for Energy Systems (old EN-509) ME-634 Thermodynamics for Energy Systems MI-101 Thermodynamics	E90 (p. 15) N91 (p. 15) N93 (p. 15)
ME-304 Power Plant Engineering / Principles of Energy Conversion (3-0-2-4)	ME-356 Principles of Energy Conversion	—	E79 (p. 15)
ME-306P Solid Mechanics Laboratory (3-0-0-3)	ME-311P Design Lab - 1	—	E88 (p. 15)
ME-308 Manufacturing Engineering (3-0-0-3)	—	ME-315 Manufacturing Engineering-II	N94 (p. 16)

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Course (code, title, L-T-P-C)	Equivalent codes (only one may be credited)	Cannot be credited together with	Rows
ME-309P Theory of Machines Lab (0-0-2-1)	—	ME-312P Design Lab - 2	N92 (p. 15)
ME-311P Design Lab - 1 (0-0-2-1)	ME-306P Solid Mechanics Laboratory	—	E88 (p. 15)
ME-312P Design Lab - 2 (0-0-2-1)	—	ME-309P Theory of Machines Lab	N92 (p. 15)
ME-315 Manufacturing Engineering-II (3-0-0-3)	—	ME-308 Manufacturing Engineering	N94 (p. 16)
ME-352 Finite Element Methods in Engineering (2.5-0.5-0-3)	CE-563 Finite Element Methods in Engineering CE-607 FEM (discontinued, 24th Senate) ME-513 Finite Element Methods in Engineering ME-641 Finite Element Method	—	E86 (p. 15)
ME-356 Principles of Energy Conversion (3-1-0-4)	ME-304 Power Plant Engineering / Principles of Energy Conversion	—	E79 (p. 15)
ME-452 Robotics and Control (Robot Kinematics, Dynamics and Control) (2-1-0-3)	AR-501 Robot Kinematics, Dynamics, and Control	—	E1 (p. 3)
ME-501 Materials Science for Failure Analysis (3-0-0-3)	ME-607 Materials Science For Failure Analysis	—	E87 (p. 15)
ME-502 (old) Functional Materials (old ME-502) (3-0-0-3)	EN-509 Functional Materials for Energy Engineering ME-609 Functional Materials MT-205 Functional Properties of Materials	—	E52 (p. 10)
ME-502 (old) Nanomanufacturing (old ME-502, 8th Senate) (3-0-0-3)	ME-509 Nanomanufacturing	—	E82 (p. 15)
ME-504 Numerical Methods for Engineering Computation (3-0-0-3)	CE-562 Numerical Methods for Engineering Computation	—	E80 (p. 15)
ME-506 Fundamentals of Fracture Mechanics (2.5-0.5-0-3)	CE-564 Fundamentals of Fracture Mechanics	—	E81 (p. 15)
ME-509 Nanomanufacturing (3-0-0-0)	ME-502 (old) Nanomanufacturing (old ME-502, 8th Senate)	—	E82 (p. 15)
ME-513 Finite Element Methods in Engineering (3-0-2-4)	CE-563 Finite Element Methods in Engineering CE-607 FEM (discontinued, 24th Senate) ME-352 Finite Element Methods in Engineering ME-641 Finite Element Method	—	E86 (p. 15)
ME-522 High-Performance Scientific Computing (2-0-1-3)	—	CS-508 Introduction to Heterogeneous Computing	N89 (p. 15)
ME-530 Continuum Mechanics (3-0-0-3)	CE-530 Continuum Mechanics	IC-242 Continuum Mechanics	E83 (p. 15) N72 (p. 12)
ME-600 Research Methodology (1-0-0-1)	BY-600 Research Methodology CE-600 Research Methodology for Civil Engineering CY-600 Research Methodology for Chemistry HC-600 Research Methodology HS-600 Research Methodology IK-509 Research methodology MA-600 Research Methodology PH-600 Research Methodology RM-510 Research Methodology RM-600 Research Methodology SC-600 Research Methodology	—	E99 (p. 17)
ME-607 Materials Science For Failure Analysis (3-0-0-3)	ME-501 Materials Science for Failure Analysis	—	E87 (p. 15)
ME-609 Functional Materials (3-0-0-3)	EN-509 Functional Materials for Energy Engineering ME-502 (old) Functional Materials (old ME-502) MT-205 Functional Properties of Materials	—	E52 (p. 10)

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Course (code, title, L-T-P-C)	Equivalent codes (only one may be credited)	Cannot be credited together with	Rows
ME-610 Advanced Thermodynamics (3-1-0-4)	—	EN-509 (old) Thermodynamics for Energy Systems (old EN-509) ME-634 Thermodynamics for Energy Systems	N90 (p. 15)
ME-617 Mechanics of Composite Materials (3-0-0-3)	CE-617 Mechanics of Composite Materials	—	E84 (p. 15)
ME-620 Modeling and Simulation (2-0-2-3)	—	EN-605 Modeling and Simulation of Energy Systems	N64 (p. 11)
ME-633 Design of Energy Systems (3-0-0-3)	EN-506 Design of Energy Systems	—	E91 (p. 15)
ME-634 Thermodynamics for Energy Systems (3-0-0-3)	EN-509 (old) Thermodynamics for Energy Systems (old EN-509)	IC-142 Engineering Thermodynamics ME-213 Engineering Thermodynamics ME-610 Advanced Thermodynamics	E85 (p. 15) N90 (p. 15) N93 (p. 15)
ME-641 Finite Element Method (3-0-0-3)	CE-563 Finite Element Methods in Engineering CE-607 FEM (discontinued, 24th Senate) ME-352 Finite Element Methods in Engineering ME-513 Finite Element Methods in Engineering	—	E86 (p. 15)
MI-101 Thermodynamics (2-1-0-3)	—	IC-142 Engineering Thermodynamics ME-213 Engineering Thermodynamics	N91 (p. 15)
MT-203 Materials Synthesis & Characterization (3-0-2-4)	—	EN-512 Structure-Property Correlations for Energy Applications	N95 (p. 16)
MT-205 Functional Properties of Materials (3-0-2-4)	EN-509 Functional Materials for Energy Engineering ME-502 (old) Functional Materials (old ME-502) ME-609 Functional Materials	—	E52 (p. 10)
MT-304 Mechanical Behavior of Materials (3-0-2-4)	—	EM-658 Deformation Behavior of Energy Materials EN-611 Durability Behavior of Energy Materials	N65 (p. 11)
MT-501 Energy Conversion & Storage Technology (3-0-0-3)	—	EN-503 Energy Storage Technologies	N96 (p. 16)
MT-506 Biomaterials (3-0-0-3)	—	BE-310 Biomaterials BE-504 Biomaterials	N5 (p. 3)
MT-509 Hydrogen Energy (3-0-0-3)	—	CY-552 Hydrogen Generation and Storage	N97 (p. 16)
MT-510 Colloids and Interfaces (3-0-0-3)	—	CY-523 Colloids and Interface Science and Technology	N31 (p. 7)
PH-301 Quantum mechanics and applications (2.5-0.5-0-3)	QS-302 Foundation of Quantum Technology	PH-513 Quantum Mechanics I	E92 (p. 16) N99 (p. 16)
PH-302 Introduction to Statistical Mechanics (2.5-0.5-0-3)	PH-422 Statistical Mechanics PH-522 Statistical Mechanics	—	E97 (p. 16)
PH-421 Electromagnetic Theory (4-0-0-4)	PH-521 Electromagnetic Theory	EE-205 Electromagnetics and Wave Propagation	E95 (p. 16) N40 (p. 9)
PH-422 Statistical Mechanics (4-0-0-4)	PH-302 Introduction to Statistical Mechanics PH-522 Statistical Mechanics	—	E97 (p. 16)
PH-424 Atomic and Molecular Spectroscopy (4-0-0-4)	—	EP-403 Physics of atoms and molecules PH-524 Atomic and Molecular Physics	N101 (p. 17) N103 (p. 17)
PH-511 Mathematical Physics (4-0-0-4)	—	EP-301 Engineering Mathematics-2	N100 (p. 17)
PH-513 Quantum Mechanics I (3-0-0-3)	—	PH-301 Quantum mechanics and applications QS-302 Foundation of Quantum Technology	N99 (p. 16)
PH-521 Electromagnetic Theory (4-0-0-4)	PH-421 Electromagnetic Theory	EE-205 Electromagnetics and Wave Propagation	E95 (p. 16) N40 (p. 9)
PH-522 Statistical Mechanics (4-0-0-4)	PH-302 Introduction to Statistical Mechanics PH-422 Statistical Mechanics	—	E97 (p. 16)
PH-524 Atomic and Molecular Physics (3-0-0-3)	—	EP-403 Physics of atoms and molecules PH-424 Atomic and Molecular Spectroscopy	N98 (p. 16) N101 (p. 17)

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Course (code, title, L-T-P-C)	Equivalent codes (only one may be credited)	Cannot be credited together with	Rows
PH-550 Introduction to Quantum Optics (3-0-0-3)	QT-511 Introduction to Quantum Optics	—	E93 (p. 16)
PH-579 Quantum Computation and Information (3-0-0-3)	—	CS-520 Introduction to Quantum Computing QT-405 Introduction to Quantum Computation	N24 (p. 6)
PH-600 Research Methodology (1-0-0-1)	BY-600 Research Methodology CE-600 Research Methodology for Civil Engineering CY-600 Research Methodology for Chemistry HC-600 Research Methodology HS-600 Research Methodology IK-509 Research methodology MA-600 Research Methodology ME-600 Research Methodology RM-510 Research Methodology RM-600 Research Methodology SC-600 Research Methodology	—	E99 (p. 17)
PH-611 Nuclear and Particle Physics (4-0-0-4)	PH-612 Nuclear and Particle Physics	—	E96 (p. 16)
PH-611P Experimental Research Techniques (0-0-7-4)	PH-614P Experimental Research Techniques	—	E94 (p. 16)
PH-612 Nuclear and Particle Physics (3-0-0-3)	PH-611 Nuclear and Particle Physics	—	E96 (p. 16)
PH-613 Quantum Mechanics II (4-0-0-4)	—	PH-702 Advanced Quantum Mechanics	N102 (p. 17)
PH-614P Experimental Research Techniques (0-0-7-4)	PH-611P Experimental Research Techniques	—	E94 (p. 16)
PH-702 Advanced Quantum Mechanics (3-0-0-3)	—	PH-613 Quantum Mechanics II	N102 (p. 17)
QS-302 Foundation of Quantum Technology (3-0-0-3)	PH-301 Quantum mechanics and applications	PH-513 Quantum Mechanics I	E92 (p. 16) N99 (p. 16)
QT-405 Introduction to Quantum Computation (3-0-0-3)	CS-520 Introduction to Quantum Computing	PH-579 Quantum Computation and Information	E17 (p. 5) N24 (p. 6)
QT-511 Introduction to Quantum Optics (3-0-0-3)	PH-550 Introduction to Quantum Optics	—	E93 (p. 16)
RM-510 Research Methodology (1-0-0-1)	BY-600 Research Methodology CE-600 Research Methodology for Civil Engineering CY-600 Research Methodology for Chemistry HC-600 Research Methodology HS-600 Research Methodology IK-509 Research methodology MA-600 Research Methodology ME-600 Research Methodology PH-600 Research Methodology RM-600 Research Methodology SC-600 Research Methodology	—	E99 (p. 17)
RM-600 Research Methodology (1-0-0-1)	BY-600 Research Methodology CE-600 Research Methodology for Civil Engineering CY-600 Research Methodology for Chemistry HC-600 Research Methodology HS-600 Research Methodology IK-509 Research methodology MA-600 Research Methodology ME-600 Research Methodology PH-600 Research Methodology RM-510 Research Methodology SC-600 Research Methodology	—	E99 (p. 17)

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Course (code, title, L-T-P-C)	Equivalent codes (only one may be credited)	Cannot be credited together with	Rows
SC-600 Research Methodology (1 cr.)	BY-600 Research Methodology CE-600 Research Methodology for Civil Engineering CY-600 Research Methodology for Chemistry HC-600 Research Methodology HS-600 Research Methodology IK-509 Research methodology MA-600 Research Methodology ME-600 Research Methodology PH-600 Research Methodology RM-510 Research Methodology RM-600 Research Methodology	—	E99 (p. 17)
VL-326 Computer Organization (3-0-2-4)	CS-201 Computer Organization CS-214 Computer Organization EE-306 Computer Organization & Microprocessor	—	E24 (p. 5)
VL-404 CMOS Analog IC Design (3-0-2-4)	EE-512 CMOS Analog IC Design	—	E48 (p. 9)

A Course records that need correction

The Mutual Exclusion field of these records names a course that does not fit. These entries are not used in this booklet.

Course	Field says	Why it looks wrong
EE-314 Digital Signal Processing (3-0-2-4)	“EEXXX (Signals & Systems)”	A placeholder code. Signals and Systems (IC-260/EE-260) is the foundation for DSP, so this reads like a prerequisite typed in the wrong field. The same text appears in EE-574 and EE-575.
EE-574 Biomedical Signal and Image Analysis (3-0-0-3)	“EEXXX (Signals & Systems)”	Same placeholder as EE-314; no course code to enforce.
EE-575 Applied Statistics for Data and Signals (3-0-2-4)	“EEXXX (Signals & Systems)”	Same placeholder as EE-314. EE-575 has the same title as EE-581; see the similarity section.
MB-202 Microeconomics (3-0-0-3)	HS-304 Organizational Management	Text is identical to MB-201’s entry. Microeconomics has no content link with Organizational Management; likely copied.
MB-550 Artificial Intelligence for Marketing (2-0-0-2)	HS-304 Organizational Management	Identical to MB-201’s entry. AI for Marketing has no content link with Organizational Management; likely copied.
MA-576 Nonlinear Optimization (3-0-2-4)	DS-413, CS-669	Identical to MA-574’s entry. Nonlinear Optimization does not overlap with statistical learning/pattern recognition; the related courses are the optimization courses (see similarity section).
MT-206 Extraction and Materials Processing (3-0-2-4)	EN-509 Functional Materials for Energy Engineering	Identical to MT-205’s entry. Extraction and Materials Processing does not match Functional Materials; likely copied.
MT-510 Colloids and Interfaces (3-0-0-3)	“Some similarity with CY552: Hydrogen Generation”	Identical to MT-509’s entry. Colloids and Interfaces matches CY-523 Colloids and Interface Science and Technology instead (see similarity section).
CS-520 Introduction to Quantum Computing (2-0-1-3)	“Other Research Methodology Courses”	Introduction to Quantum Computing is not a Research Methodology course; the text was copied from the RM template.
CY-524 Basic and Applied Electrochemistry EE-542 Modelling, Simulation and Control of Hybrid Electric Vehicle IC-141 Product Realization Technology IC-181 Introduction to Consciousness and Holistic Wellbeing MB-531 Ethical and Legal aspects of Business MB-592 Management Science In Practice – A Modelling And Case Studies Approach With Ms-Excel. ME-100 Reverse Engineering ME-511 Manufacturing of Composites	template sentence only	The field holds the proposal template’s instruction (“courses with high similarity not allowed to credit ...”) and names no course.
CS-580 Advanced Data Structures and Algorithms	(code conflict)	The DC Curriculum treats CS-580 as the earlier code of CS-514 (used in this booklet). The master-list CS-580 record, however, carries an old B.Tech syllabus close to CS-202. Please confirm.

B L-T-P-C values to be confirmed

All 350 courses in this booklet were checked against the Master Course List, the course proposal and the DC Curriculum, and against $C = L + T + P/2$. 304 are consistent. The 46 below need confirmation; the booklet shows the DC value where there is one.

Course	Title	Shown	Master list	Proposal	DC	Observation
BE-304	Bioinformatics	3-0-2-4	2-0-2-3	3-0-2-4	3-0-2-4	sources differ
CE-302P	Geotechnical Engineering Lab	L-T-P-C not recorded	—	—	—	L-T-P-C not recorded
CE-607	FEM (discontinued, 24th Senate)	L-T-P-C not recorded	—	—	—	L-T-P-C not recorded
CS-207	Applied Databases Practicum	0-0-3-2	0-0-3-2	—	(2 cr.)	$C \neq L+T+P/2$
CS-312	Operating System	3-0-2-4	3-0-0-3	—	3-0-2-4	sources differ
CS-403	Algorithm Design and Analysis	3-0-2-4	3-0-2-4	—	(3 cr.)	DC credits 3
CS-508	Introduction to Heterogeneous Computing	2-0-0-2	2-0-0-2	—	(3 cr.)	DC credits 3
CS-520	Introduction to Quantum Computing	2-0-1-3	2-0-1-3	—	(3 cr.)	$C \neq L+T+P/2$

Course	Title	Shown	Master list	Proposal	DC	Observation
CS-580	Advanced Data Structures and Algorithms (old)	3-0-1-4	3-0-1-4	—	—	$C \neq L+T+P/2$
CS-609	Speech Processing	3-0-2-4	3-0-2-4	—	(3 cr.)	DC credits 3
CS-683	Generative AI	3-0-1-4	3-0-1-4	—	(4 cr.)	$C \neq L+T+P/2$
CY-531P	Organic Chemistry Laboratory	3 cr.	—	—	(3 cr.)	L-T-P-C not recorded; credits 3 only
CY-532	Photochemistry and Pericyclic Reactions	3 cr.	—	—	(3 cr.)	L-T-P-C not recorded; credits 3 only
CY-533	Chemistry of Main Group Elements	3 cr.	—	—	(3 cr.)	L-T-P-C not recorded; credits 3 only
CY-533P	Inorganic Chemistry Lab	0-0-8-4	0-0-8-4	—	(3 cr.)	DC credits 3
CY-534	Chemistry of Transition Elements	3-1-0-4	3-1-0-4	—	(3 cr.)	DC credits 3
EE-101	Electrical Science	3-1-2-4	3-1-2-4	—	(4 cr.)	$C \neq L+T+P/2$
EE-205	Electromagnetics and Wave Propagation	3-0-0-3	2.5-0.5-0-0	—	3-0-0-3	sources differ
EE-302	Control Systems	3-0-2-4	3-0-0-3	—	3-0-2-4	sources differ
EE-304	Communication Systems	3-0-1-4	3-0-2-4	—	3-0-1-4	sources differ; $C \neq L+T+P/2$
EE-533	Access Network	4 cr.	—	—	(4 cr.)	L-T-P-C not recorded; credits 4 only
EE-604	Advanced Electric Drives	3 cr.	—	—	(3 cr.)	L-T-P-C not recorded; credits 3 only
EE-605	Information Theory	3 cr.	—	—	(3 cr.)	L-T-P-C not recorded; credits 3 only
EE-677	Analog Circuit Design	3 cr.	—	—	(3 cr.)	L-T-P-C not recorded; credits 3 only
EE-678	Digital Circuit Design	3 cr.	—	—	(3 cr.)	L-T-P-C not recorded; credits 3 only
EM-601	(title not recorded)	L-T-P-C not recorded	—	—	—	L-T-P-C not recorded
IC-140	Engineering Graphics for Design	2-0-3-4	2-0-3-4	—	2-0-3-4	$C \neq L+T+P/2$
IK-101	Sanskrit Language – Level I (UG)	L-T-P-C not recorded	—	—	—	L-T-P-C not recorded
IK-593	Philosophy of Science	3 cr.	—	—	—	L-T-P-C not recorded; credits 3 only
MA-221	Numerical Analysis	3-1-0-4	2-0-3-4	—	3-1-0-4	sources differ
MA-323	Applied Databases Practicum	0-0-3-2	0-0-3-2	—	—	$C \neq L+T+P/2$
MA-549	Abstract Algebra	3-0-0-3	3-0-0-3	—	(4 cr.)	DC credits 4
MA-665	Semigroup Of Bounded Linear Operators	3 cr.	—	—	(3 cr.)	L-T-P-C not recorded; credits 3 only
MB-201	Foundations of Business Management	3-0-1-4	3-0-1-4	—	—	$C \neq L+T+P/2$
MB-203	Macroeconomics	3 cr.	—	—	—	L-T-P-C not recorded; credits 3 only
MB-303	Financial Accounting	3 cr.	—	—	—	L-T-P-C not recorded; credits 3 only
MB-402	Operations Research	3 cr.	—	—	—	L-T-P-C not recorded; credits 3 only
ME-212	Product Manufacturing Technology	2-0-3-3	2-0-2-3	—	2-0-3-3	sources differ; $C \neq L+T+P/2$
ME-452	Robotics and Control (Robot Kinematics, Dynamics and Control)	2-1-0-3	2-1-0-3	—	(4 cr.)	DC credits 4
ME-509	Nanomanufacturing	3-0-0-0	3-0-0-3	3-0-0-0	(3 cr.)	sources differ; $C \neq L+T+P/2$; DC credits 3
ME-522	High-Performance Scientific Computing	2-0-1-3	2-0-1-3	2-0-1-3	(3 cr.)	$C \neq L+T+P/2$
PH-513	Quantum Mechanics I	3-0-0-3	4-0-0-4	3-0-0-3	(3 cr.)	sources differ
PH-611P	Experimental Research Techniques	0-0-7-4	0-0-7-4	—	—	$C \neq L+T+P/2$
PH-613	Quantum Mechanics II	4-0-0-4	4-0-0-4	—	(3 cr.)	DC credits 3
PH-614P	Experimental Research Techniques	0-0-7-4	0-0-7-4	—	—	$C \neq L+T+P/2$
SC-600	Research Methodology	1 cr.	—	—	—	L-T-P-C not recorded; credits 1 only

C Other restrictions

These apply to a programme or a minor, not to a pair of courses.

Course(s)	Title	Restriction
DS-302 / DS-312	Computing Systems for Data Processing	“Not allowed to B.Tech. CSE” (distribution line in the DS-302 record).
CS-515	Advanced Computer Science Practicum	Core for M.Tech (CSE); “B.Tech students will not be allowed to credit the same” because its content overlaps several B.Tech core courses (CS-207, CS-309, CS-310, CS-502).
DS-417 / DS-418	Deep Learning / Introduction to Generative AI	DSAI-only codes; students of other programmes take CS-671 / CS-683 instead (DC curriculum note).
HS-510 / HS-504	Essentials of Entrepreneurship / Personal Finance and Portfolio Management	Minor in Management: “only one course from each pair may be counted” toward the minor. The two are different courses, so this is a minor-counting rule, not a general exclusion.



Course(s)	Title	Restriction
QT-302 / PH-513 / EP-301, QT-408 / PH-532 / PH-501, QT-510 / PH-601, QS-501P / QT-304P	Quantum Technologies minor	Listed on one line in the Quantum minor as alternatives that satisfy one requirement. The master-list titles of the PH/EP codes differ from the QT titles (e.g. EP-301 is Engineering Mathematics-2, PH-601 is Mesoscopic Physics), so they are treated as alternatives, not as the same course.